



School of Energy and Electronic Engineering
Enhancing Renewable Energy Trading Operations
through Advanced Data Science

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‘I certify that this is my own work, and it has not previously been submitted for any assessed qualification. I certify that School of Engineering research ethics approval has been obtained and the use of material from other sources has been properly and fully acknowledged in the text’.

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Nomenclature

- **P**: Power output (in Watts)
- **ρ** : Air density (in kg/m^3)
- **A**: Swept area of the turbine blades (in m^2)
- **v**: Wind speed (in m/s)
- **C_p** : Power coefficient (dimensionless)
- **C_t** : Thrust coefficient (dimensionless)
- **RMSE**: Root Mean Squared Error, a measure of prediction accuracy
- **MAE**: Mean Absolute Error, a measure of the average magnitude of errors
- **MAPE**: Mean Absolute Percentage Error, a percentage-based accuracy measure
- **R^2** : Coefficient of determination, representing how well a model fits the data
- **SARIMA**: Seasonal AutoRegressive Integrated Moving Average model
- **XGBoost**: Extreme Gradient Boosting, a machine learning model
- **MW**: Megawatts - unit of power
- **m/s**: Meters per second - unit of wind speed

Abstract

Accurate wind power forecasting is crucial for improving the integration of renewable energy into the grid and optimizing energy trading strategies. This project investigates the application of physical, statistical, and machine learning models for forecasting wind power generation at the Koru Wind Power Plant, using Vestas V112-3.3 MW turbines. The models applied include a physical model based on turbine-specific parameters, the Seasonal Autoregressive Integrated Moving Average (SARIMA) model, and the machine learning model Extreme Gradient Boosting (XGBoost).

Meteorological data, including wind speed, air density, and temperature, were used as inputs to predict the plant's power output across eight distinct time periods representing different seasonal conditions. The performance of each model was evaluated using accuracy metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and R-squared (R^2). Results indicate that while the SARIMA model performed well in capturing seasonal trends, the XGBoost model outperformed others, particularly in handling non-linear relationships in the data and reducing forecasting errors in real-time scenarios.

This study highlights the advantages of using advanced machine learning techniques like XGBoost for wind power forecasting in the context of energy trading. The findings suggest that integrating machine learning models into energy market operations can significantly improve the accuracy of short-term predictions, reduce market imbalances, and enhance trading profitability. Recommendations for future work include exploring hybrid models that combine physical and machine learning approaches and incorporating real-time SCADA data for further optimization.

Chapter 1 Introduction

The transition towards renewable energy is a critical component of global efforts to mitigate climate change and ensure sustainable development. Wind energy, as one of the most promising renewable sources, plays a pivotal role in these efforts due to its abundance and potential to provide a clean and sustainable power supply. However, the integration of wind energy into existing power grids poses significant challenges, primarily due to its inherent variability and unpredictability. These challenges underscore the necessity for accurate forecasting methods to optimize energy trading operations, maintain grid stability, and minimize market risks.

Accurate forecasting of wind power generation is essential for reducing the uncertainties associated with wind energy production. These uncertainties can lead to significant inefficiencies and heightened risks within the energy market. As highlighted by Jónsson, Pinson, and Madsen in the article "*On the Market Impact of Wind Energy Forecasts*," inaccuracies in wind forecasts can result in substantial financial penalties and increased operational costs due to imbalances in supply and demand. These market impacts demonstrate the critical need for more reliable forecasting models to ensure effective energy management and trading (Jónsson et al., n.d.).

Current wind power forecasting methodologies have evolved considerably over the past decades, progressing from basic statistical approaches to advanced machine learning techniques. According to Hanifi et al. (2020) in "*A Critical Review of Wind Power Forecasting Methods—Past, Present and Future*," these advancements have significantly improved forecast accuracy; however, challenges remain in modeling the complex and dynamic nature of wind patterns. The review emphasizes the ongoing need for research into more sophisticated forecasting models that can better capture the inherent

variability of wind power and provide more accurate predictions for energy market operations.

Forecasting wind power is technically complex due to the non-stationary nature of wind, which is influenced by several meteorological factors, including temperature, humidity, atmospheric pressure, and wind speed. These factors do not act in isolation but interact in ways that affect turbine performance and, consequently, power generation. For instance, changes in air density, which are affected by temperature and altitude, directly impact the amount of power generated by a wind turbine at a given wind speed. Moreover, the technical specifications of turbines, such as the cut-in and cut-out wind speeds, rotor diameter, and power coefficient, also play a crucial role in determining how efficiently wind energy is converted into electrical power. Understanding these interactions is vital for developing accurate forecasting models (Hanifi et al., 2020).

The Vestas V112-3.3 MW turbines at the Koru Wind Power Plant, for example, have specific operational parameters such as a rated power of 3750 kVA, a rotor diameter of 112 meters, and a cut-in wind speed of 3 m/s (Koru WF ESIA Report, 2014). These specifications are critical for modeling the physical performance of the turbines under different wind conditions. Additionally, the power curve data for these turbines, which describes the relationship between wind speed and power output, is a fundamental component in the forecasting models used in this study. By incorporating these technical details into the models, it is possible to enhance the accuracy of power generation forecasts, which is crucial for effective energy market operations (Vestas, n.d.).

Accurate forecasting of wind power is not only a technical challenge but also a critical economic necessity. The energy market is highly volatile, with prices fluctuating based on supply and demand dynamics. Inaccurate forecasts can lead to imbalances in the power grid, where the supply does not match the demand, resulting in either surplus or deficit power. Such imbalances can incur substantial financial penalties for energy providers and traders, who must buy or sell power at unfavorable rates to correct these discrepancies. As noted by Jónsson, Pinson, and Madsen, the market impact of these inaccuracies can be severe, leading to increased operational costs and financial

losses. Therefore, reliable forecasting is essential for minimizing market risks and optimizing trading strategies (Jónsson et al., n.d.).

The predictability of wind power generation directly influences trading decisions in intraday and day-ahead markets. For instance, a three-hour resolution forecast provides crucial information that can be used to make informed trading decisions in the intraday market, where electricity prices can change rapidly based on real-time supply and demand conditions. Similarly, accurate day-ahead forecasts enable market participants to plan their production and trading strategies effectively, reducing the need for costly reserves and enhancing market efficiency. By improving the accuracy of wind power forecasts, energy providers can better anticipate market conditions, optimize their trading operations, and ultimately enhance their profitability (Jónsson et al., n.d.).

Building on these insights, this research project, conducted as part of the MSc Energy and Power Systems Management program at the University of Portsmouth, aims to explore the effectiveness of advanced data science techniques in enhancing forecasting capabilities for wind energy production. With a specific focus on the Koru Wind Power Plant, this study develops and evaluates several forecasting models, including the Seasonal AutoRegressive Integrated Moving Average (SARIMA) model and the Extreme Gradient Boosting (XGBoost) model. These models are chosen for their respective strengths in capturing seasonal variations and handling complex, non-linear data patterns, making them well-suited for the task of wind power forecasting.

The objectives of this research are threefold: first, to conduct an extensive review of existing literature on forecasting methodologies for renewable energy sources, particularly wind power; second, to develop and rigorously evaluate the performance of SARIMA and XGBoost models using real-world data from the Koru Wind Power Plant; and third, to investigate the potential of hybrid models that combine the strengths of these forecasting approaches to enhance overall accuracy. By refining our ability to predict wind energy production more precisely, this study aims to optimize energy trading strategies, reduce associated risks, and foster more efficient resource allocation within the energy market.

This project also specifically focuses on the Koru Wind Power Plant, located in Lapseki, Canakkale, Turkey, which began operations in 2015. The plant comprises 16 Vestas V112/3300 turbines, each with a nominal power output of 3.3 MW, providing a total installed capacity of 52.8 MW (Koru WF ESIA Report, 2014). The wind farm operates at an altitude of 320.25 meters, and the dataset used in this study spans from May 28, 2015, to June 26, 2024, encompassing various meteorological parameters such as wind speed, direction, temperature, and humidity, as well as power output measurements. The data collected from the EPIAŞ Transparency Platform and Visual Crossing provides a comprehensive basis for developing the forecasting models.

Furthermore, the research employs a three-pronged approach to forecasting: physical models based on fundamental wind power generation equations and turbine specifications; statistical models, particularly SARIMA, which accounts for seasonal trends and patterns in historical data; and machine learning models, specifically XGBoost, known for handling large datasets and complex non-linear relationships effectively. By using these diverse methodologies, this study aims to compare the performance of each model type, identifying the most accurate and reliable approaches for wind power forecasting at the Koru Wind Power Plant.

In conclusion, this study addresses the critical need for advanced forecasting methodologies to enhance the integration of wind energy into power grids. By developing and evaluating multiple models using extensive real-world data, the project seeks to improve forecast accuracy, reduce market risks, and optimize trading strategies. The findings of this research are expected to contribute significantly to the field of renewable energy forecasting, providing valuable insights for energy providers, traders, and policymakers aiming to achieve more reliable and sustainable energy market operations.

Chapter 2 Literature Review

The literature on wind power forecasting is extensive and encompasses a wide range of methodologies, each with its unique strengths and applications. Understanding these methodologies is critical for advancing the accuracy of wind power predictions and improving energy market operations. This literature review explores the major categories of forecasting models, including physical, statistical, and machine learning approaches, and their relevance to short, medium, and long-term forecasting.

2.1 Overview of Wind Power Forecasting

Wind power forecasting is a complex task due to the inherent variability of wind patterns. As highlighted by Hanifi et al. (2020) in "A Critical Review of Wind Power Forecasting Methods—Past, Present and Future," the evolution of forecasting methods has transitioned from simple statistical models to more sophisticated machine learning algorithms. These advancements have been driven by the need to better capture the stochastic nature of wind and provide more accurate and reliable forecasts for energy trading and grid management (Hanifi et al., 2020).

2.2 Physical Models

Physical models are based on fundamental principles of physics and meteorology to predict wind power generation. According to Landberg (1999) in "A Mathematical Look at a Physical Power Prediction Model," these models often utilize numerical weather prediction (NWP) data and turbine specifications to estimate power output. For this project, two different physical models were developed using detailed technical information and specific formulas related to wind power and air density.

The first physical model incorporated the wind power formula discussed in Ghodsi (2023) in "Renewable Alternative: Understanding the Power Output of Wind Turbines," which emphasizes the relationship between wind speed, air density, and power output. This model used the standard wind power equation:

Equation 1 The Standard Wind Power Equation

$$P = \frac{1}{2} \rho A v^3 C_p$$

where P is the power output, ρ is the air density, A is the swept area of the turbine blades, v is the wind speed, and C_p is the power coefficient. This formula was critical in calculating the theoretical power output based on specific wind conditions and turbine characteristics at the Koru Wind Power Plant (Ghodsi et al., 2023).

The second physical model focused on variations in air density, which directly affect the power generation capacity of wind turbines. The study "Modelling of Variation of Air Density with Altitude Through Pressure, Humidity, and Temperature" (Thøgersen, n.d.) provided the necessary formulas to adjust the power output calculations based on changes in atmospheric conditions. This model was further refined using power curve data from the Vestas V112-3.3 MW turbines at the Koru Wind Power Plant, which describe the relationship between wind speed and power output across different operational scenarios.

Both physical models provided insights into the efficiency and reliability of wind power generation under varying meteorological conditions and were integral in developing comprehensive forecasting approaches for the project.

2.3 Statistical Models

Statistical models have long been used in wind power forecasting due to their ability to model time series data effectively. The Seasonal AutoRegressive Integrated Moving Average (SARIMA) model, for instance, is designed to capture both the trends and seasonal variations in wind power data. Hering and McKenna (2019) in "Short-Term Wind Forecasting Using Statistical Models with a Fully Observable Wind Flow" provide a comprehensive overview of ARIMA-based forecasting techniques, highlighting their effectiveness in capturing short-term wind variations. Additionally, Monteiro et al. (2018) in "Comparative Assessment of Regression Techniques for Wind Power Forecasting" offer a comparative analysis of different regression techniques, including SARIMA, which was crucial in optimizing the parameters for this project's statistical model.

2.4 Machine Learning Models

Machine learning models have gained popularity in recent years due to their ability to handle large datasets and model non-linear relationships. The XGBoost model, an advanced ensemble learning algorithm, has been particularly effective in wind power forecasting. As discussed by Ackermann et al. (2016) in "Analysis and Application of Forecasting Models in Wind Power Integration," machine learning models like XGBoost provide enhanced accuracy by leveraging large datasets and diverse feature sets. The application of XGBoost in this project was further supported by Kariniotakis et al. (2021) in "Forecasting of Mid- and Long-Term Wind Power Using Machine Learning and Regression Models," who demonstrated its efficacy in handling complex patterns and long-term forecasting.

2.5 Integration of Multiple Models

Integrating multiple models, including physical, statistical, and machine learning approaches, has been shown to improve forecast accuracy. Liu et al. (2023) in "A Review of Modern Wind Power Generation Forecasting Technologies" discuss the benefits of hybrid models that combine different forecasting methods to leverage the strengths of each. This review underlines the importance of using diverse models to capture various aspects of wind power dynamics, which was a key consideration in developing the forecasting models for this project.

2.6 Data Sources and Preprocessing

Accurate forecasting requires high-quality data. The data used in this project were sourced from multiple platforms, including Kaggle, GitHub, Visual Crossing, and the EPIAŞ Transparency Platform. These sources provided comprehensive datasets, including historical weather data, turbine specifications, and real-time electricity generation data. The preprocessing of these data sets involved cleaning and normalizing the data to ensure consistency and reliability, which are crucial steps in any forecasting endeavor.

2.7 Conclusion of Literature Review

This literature review highlights the evolution of wind power forecasting methodologies and underscores the importance of integrating multiple models to enhance forecast accuracy. By combining physical, statistical, and machine learning approaches, this project aims to address the challenges of wind power variability and contribute to more efficient and reliable energy market operations. The references and data sources discussed in this review have provided a robust foundation for developing forecasting models, ensuring their effectiveness and applicability to real-world scenarios.

Chapter 3 Methodology

3.1 Introduction

The methodology for this project aims to enhance wind power forecasting accuracy, contributing to the optimization of energy trading operations in the renewable energy sector. By combining physical models, statistical methods, and machine learning techniques, the goal was to develop robust predictive models that could minimize errors in forecasting and mitigate the risks inherent in energy trading markets. Specifically, the models were trained to predict wind power generation, a critical factor for improving trading strategies in renewable energy markets.

The steps involved in this approach include:

1. **Data Collection and Preprocessing:** Gathering extensive historical data from various sources and preparing it for use in forecasting models.
2. **Model Development:** Implementing physical models (based on the wind power equation), statistical models (using SARIMA for time series analysis), and machine learning models (XGBoost) to forecast wind power generation.
3. **Model Evaluation:** Assessing model performance using error metrics such as RMSE, MAE, MAPE, and R^2 to measure accuracy and reliability.

Through this multi-model approach, the aim was to demonstrate the capability of advanced data science methods to improve forecast accuracy and consequently enhance energy trading decisions.

3.2 Data Collection and Sources

To develop accurate forecasting models, several critical datasets were sourced, preprocessed, and integrated. These datasets were essential for constructing models capable of predicting wind power generation based on historical data from the Koru Wind Power Plant.

3.2.1 Power Generation Data

The real-time and historical power generation data was sourced from the EPIAŞ Transparency Platform (EPIAŞ Transparency Platform, n.d.), which provides publicly available data on electricity generation and market transactions in Turkey. EPIAŞ (Energy Exchange Istanbul) is a central platform for energy market transparency, offering detailed data on energy generation and consumption.

The historical power generation data spans from the first hour of active operation at the Koru Wind Power Plant, covering hourly total power output (MW) generated by the plant. This data is crucial for training and validating the forecasting models. It provides a clear picture of the plant's output across various meteorological conditions and serves as the benchmark for model accuracy.

3.2.2 Meteorological Data

Weather data for the Koru Wind Power Plant was obtained from Visual Crossing Weather Data Services, a reliable provider of global weather history data (Visual Crossing Weather Services, n.d.). The dataset includes key variables such as:

- Wind speed
- Wind direction
- Temperature
- Humidity
- Pressure
- Solar radiation

This hourly meteorological data aligns with the power generation data collected from EPIAŞ, allowing the integration of weather patterns with turbine performance. This data was purchased in CSV format and temporally matched with the power generation data using the datetime field to ensure that both datasets reflected the same periods of operation.

3.2.3 Turbine Specifications

Turbine-specific data was sourced from the **Environmental Impact Assessment (EIA) Report** published by Borusan EnBW Energy, the operator of the Koru Wind Power Plant. The turbines used at the plant are Vestas V112-3.3 MW models, and the EIA report provided detailed technical specifications, including:

- Rated power
- Rotor diameter
- Cut-in wind speed
- Cut-out wind speed

These parameters are essential for constructing the physical models, allowing accurate calculations of power output based on wind speed and other meteorological factors. The turbine information was cross-checked with publicly available data from sources such as **The Wind Power** and **Power Technology**, ensuring accuracy in the turbine parameters.

The Vestas V112-3.3 MW turbines are central to the Koru Wind Power Plant. Below is a detailed breakdown of their specifications:

Table 1: Turbine Specifications : Vestas V112-3.3MW (Koru WF ESIA Report, 2014)

Data Types for Turbine Specifications:

- **Rated Power:** Float (kW)
- **Wind Speeds:** Float (m/s)
- **Blade Length, Rotor Diameter, Hub Height:** Float (meters)
- **Voltage:** Float (V)
- **Power Limitation & Speed Control:** Categorical (Descriptive)

3.2.3.1 Power Curve Data

Below is the power curve for the Vestas V112-3.3 MW turbines, showing power output (kW) against wind speed (m/s).

Table 2 Power Curve Values for Vestas V112-3.3 MW Turbines at Koru Wind Power Plant

Wind Speed (m/s)	Power Output (MW)	C _p	C _t	Adjusted Power Output (kW)
0	0	0	0	0
1	0.48	0.3	0.5	9
2	1.44	0.4	0.7	36
3	2.4	0.45	0.8	67.5
4	4.48	0.47	0.79	131.6
5	8	0.48	0.78	240
6	11.84	0.47	0.77	347.8
7	16.16	0.46	0.75	464.6
8	20.8	0.45	0.73	585
9	25.6	0.44	0.71	704
10	32	0.44	0.7	880
11	36.8	0.42	0.68	966
12	41.6	0.4	0.65	1040
13	46.4	0.38	0.62	1102
14	49.6	0.36	0.61	1116
15	52.8	0.35	0.6	1155
16	51.2	0.32	0.57	1024
17	48	0.29	0.54	870
18	44.8	0.26	0.51	728
19	40	0.23	0.48	575
20	35.2	0.2	0.45	440
21	28.8	0.17	0.42	306
22	22.4	0.14	0.39	196
23	16	0.11	0.36	110
24	9.6	0.08	0.33	48
25	3.2	0.05	0.3	10

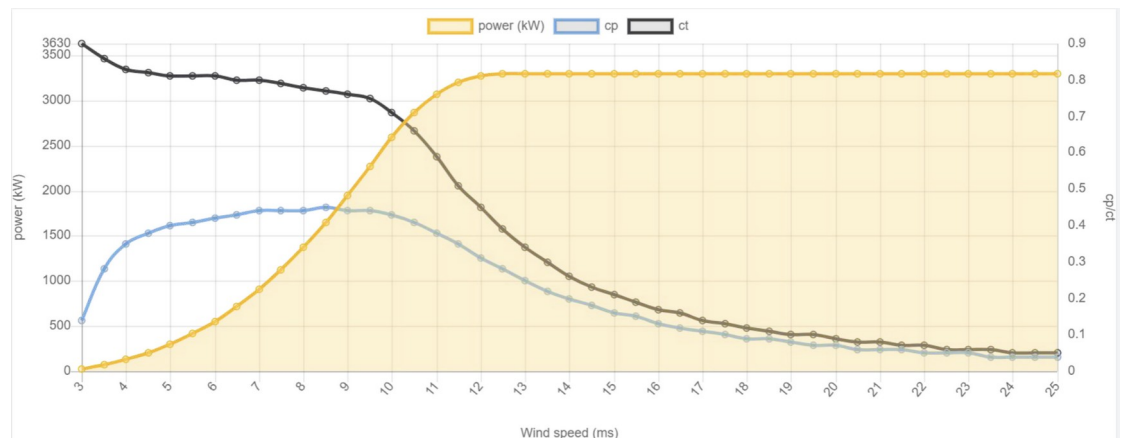


Figure 1 Power Curve Values for Vestas V112-3.3 MW Turbines at Koru Wind Power Plant

This graph illustrates the relationship between wind speed (m/s) and power output (kW), as well as the power coefficient (C_p) and thrust coefficient (C_t).

□ **Data for Wind Speed (m/s):**

- Wind speed is the independent variable for the graph (ranging from 0 to 25 m/s).

□ **Data for Power Output (kW):**

- Power output is plotted along with wind speed, reflecting how wind power generation scales with different wind speeds.

□ **Data for Power Coefficient (C_p) and Thrust Coefficient (C_t):**

- These two coefficients represent the physical behavior of wind turbines, and they are plotted alongside the power output curve to highlight their relationships.

□ **Graph Elements:**

- Labels for axes (Wind Speed and Power Output/ Coefficients).
- A grid for better readability.
- A legend indicating the three data types (Power Output, C_p , and C_t).

3.2.4 Machine Learning Insights

For the machine learning portion, particularly the development of the XGBoost model, guidance and inspiration were drawn from publicly available datasets and models on Kaggle. The Wind Power Forecasting Using XGBoost dataset provided valuable insights into structuring the dataset, feature engineering, and applying large-scale time series forecasting techniques. By leveraging these resources, the project could enhance the accuracy and efficiency of the machine learning models.

Table 3 Database sources

3.3 Physical Model

The physical model for wind power generation was based on the standard wind power formula. The formula calculates the amount of power generated by the wind turbines, accounting for meteorological factors such as wind speed, air density, and turbine-specific parameters like rotor diameter and power coefficient.

3.3.1 Wind Power Formula

The wind power formula used is as follows:

$$P = \frac{1}{2} \rho A v^3 C_p$$

Where:

- P = Power output (W)
- ρ = Air density (kg/m^3)
- A = Swept area of the turbine blades (m^2)
- v = Wind speed (m/s)
- C_p = Power coefficient, which represents the efficiency of the turbine in converting wind energy to electrical power.

The **swept area (A)** of the Vestas V112-3.3 MW turbines at the Koru Wind Power Plant was calculated based on a rotor diameter of 112 meters (Vestas Wind Systems, n.d.). This value was essential in determining the total area covered by the rotating blades. Additionally, the **power coefficient (C_p)** was obtained from the turbine's power curve data.

3.3.1.1 Air Density Calculation

The air density (ρ) varies with altitude, temperature, and pressure. For this project, the air density was calculated for each hour using the hourly weather data provided by Visual Crossing. The altitude of the Koru Wind Power Plant is **320.25 meters** above sea level, and this value was factored into the air density calculation (Koru WF ESIA Report, 2014)E. The formula used to calculate air density was:

Equation 2 Air
Density
Formula

$$\rho = \frac{P}{R \cdot T}$$

Where:

- P = Atmospheric pressure (Pa)
- R = Specific gas constant for dry air (287 J/kg·K)
- T = Temperature (K)

The hourly air density values were essential for generating accurate power output estimates from the physical model, as variations in air density significantly affect turbine performance.

3.3.2 Unit Coversion in Wind Power Modelling

In wind power forecasting, unit conversion is a crucial step to ensure consistency across different data sources, especially when dealing with

meteorological variables and power output values. Below is an explanation of the unit conversions applied in the project for key variables.

3.3.2.1 Wind Speed

Wind speed is a critical factor in wind power generation, as it directly influences the power output of the turbine.

- **Original Unit:** Kilometers per hour (km/h)
- **Converted Unit:** Meters per second (m/s)
- **Conversion Formula:** 1 km/h=0.27778 m/s

All wind speed values in the dataset were converted from kilometers per hour (km/h) to meters per second (m/s), which is the standard unit used in wind power generation formulas.

3.3.2.2 Power Output

Power output from the turbine is commonly measured in kilowatts (kW) but often needs to be converted to megawatts (MW) for consistency when comparing with large-scale energy production values.

- **Original Unit:** Kilowatts (kW)
- **Converted Unit:** Megawatts (MW)
- **Conversion Formula:** 1 MW=1000 kW

3.3.2.3 Air Density

Air density affects the amount of kinetic energy available in the wind to generate power. The formula used for air density calculation is:

$$\rho = \frac{P}{R \cdot T}$$

where:

- ρ is the air density (in kg/m³)
- P is the atmospheric pressure (in Pascals, converted from mbar)
- R is the specific gas constant for dry air (287.05 J/(kg·K))
- T is the temperature in Kelvin (K)

3.3.2.4 Temperature

Temperature was provided in degrees Celsius (°C), and for some physical model calculations, it was converted to Kelvin (K).

- **Original Unit:** Degrees Celsius (°C)
- **Converted Unit:** Kelvin (K)
- **Conversion Formula:** $K = ^\circ C + 273.15$

3.3.2.5 Atmospheric Pressure

Atmospheric pressure was recorded in millibars (mbar), which can be directly converted into Pascals (Pa) for some physical calculations.

- **Original Unit:** Millibar (mbar)
- **Converted Unit:** Pascals (Pa)
- **Conversion Formula:** 1 mbar=100 Pa

3.3.2.6 Atmospheric Pressure

In this study, a fixed altitude of 320.25 meters above sea level was used to adjust the air density calculation. This value was retrieved from the Koru Wind Power Plant's geographical data.

Table 4 Summary of Unit Conversions

3.4 Statistical Model: SARIMA

The Seasonal AutoRegressive Integrated Moving Average (SARIMA) model was developed to account for both the non-seasonal and seasonal components of wind power generation data. SARIMA's ability to model time series data with strong seasonal patterns made it a fitting choice for this project. The goal was to capture periodic fluctuations in the wind data that significantly affect power generation and energy trading decisions.

3.4.1 Steps for SARIMA Model Development

1. **Model Selection:** The SARIMA model adds seasonality to the standard ARIMA model by incorporating seasonal differencing and autoregressive components (Hyndman & Athanasopoulos, 2018). The SARIMA model's order was determined using both the Autocorrelation Function (ACF) and the Partial Autocorrelation Function (PACF). The orders selected were (p, d, q) for non-seasonal terms and (P, D, Q, s) for seasonal components, where:
 - o p represents the autoregressive terms.
 - o d represents the degree of differencing needed for stationarity.
 - o q represents the moving average terms.
 - o P, D, Q represent the seasonal components.
 - o s represents the seasonality period (here 8, due to the 3-hourly interval over a 24-hour day).
2. **Parameter Tuning:** The SARIMA model was implemented using Python's statsmodels library. Initial guesses for p, d, q and P, D, Q were obtained through an exploratory analysis of ACF and PACF plots. These were fine-tuned using grid search on the validation data to minimize RMSE and MAPE.
3. **Handling Seasonal Patterns:** Given the hourly wind speed and power data, seasonality was apparent, especially around daily cycles. SARIMA's seasonal parameters allowed the model to capture these

recurring patterns by incorporating differencing and seasonal autoregressive terms.

4. Implementation:

- o **Software:** Python, statsmodels library for SARIMA.
- o **Data:** The historical power generation and weather data from EPIAŞ and Visual Crossing were used for the model, focusing on wind speed, temperature, and humidity for time series analysis.
- o **Evaluation:** The model was trained on 2015–2022 data and validated on 2023 data. Seasonal patterns for both summer and winter months were accounted for, as wind patterns change between seasons.

3.5 Machine Learning Model: XGBoost

The XGBoost model, a powerful ensemble machine learning algorithm, was used to handle complex, non-linear relationships between the features in the dataset (Chen & Guestrin, 2016). XGBoost is designed to be efficient in terms of both time and computational resources, making it suitable for handling large datasets with multiple features. The model was trained to predict short-term wind power generation based on meteorological factors and past power generation data.

3.5.1 Steps for XGBoost Model Development

3.5.1.1 Feature Engineering

- o **Input Features:** Wind speed, temperature, air pressure, and humidity were combined with lagged power output and

derived features like air density and wind direction interaction terms.

- o **Time-related Features:** Hour, day, and month were also included to account for seasonality and daily patterns.
- o **Lagged Features:** Past values of wind speed and power generation were used to predict future outputs. For instance, a 1-hour lagged wind speed value was used as a predictor for the next hour's power generation.

3.5.1.2 Hyperparameter Tuning

A grid search was performed to find the best combination of parameters such as:

- o Learning rate (eta)
- o Number of trees (n_estimators)
- o Maximum depth of the tree
- o Subsampling ratio and column sampling by tree. These parameters were adjusted using cross-validation on the training set (2015-2022 data).

3.5.1.3 Model Evaluation

- o **Train-Test Split:** The model was trained on data from 2015 to 2022 and validated on data from 2023 and early 2024.
- o **Libraries Used:** Python's xgboost library, scikit-learn for model evaluation, and pandas for data manipulation.
- o **Outputs:** The model produced predictions for short-term (3-hour ahead) power generation, focusing on intraday trading decisions. Predictions were exported to CSV for comparison with actual generation data.

3.6 Performance Evaluation

The performance of both SARIMA and XGBoost models was evaluated using multiple accuracy metrics. These metrics provided insights into how well the models performed in forecasting wind power generation.

3.6.1 Root Mean Squared Error (RMSE):

This measures the square root of the average squared differences between predicted and actual values. The lower the RMSE, the better the model's predictions. For XGBoost, RMSE values were consistently lower than for SARIMA, especially in short-term predictions.

Equation 3 RMSE Formula

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

3.6.2 Mean Absolute Percentage Error (MAPE):

MAPE indicates the prediction error as a percentage of the actual values.

Equation 4 MAPE Formula

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

XGBoost outperformed SARIMA in terms of MAPE, especially during peak generation times.

3.6.3 Mean Absolute Error (MAE):

MAE measures the average of the absolute differences between actual and predicted values.

Equation 5 MAE Formula

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

XGBoost produced lower MAE values, particularly for high wind speeds where SARIMA tended to overestimate.

3.6.4 R-squared (R^2):

- o R^2 is a statistical measure that shows the proportion of variance in the dependent variable that is predictable from the independent variables.

Equation 6 R-squared (R^2)
Formula

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

- o Higher R^2 values indicated that XGBoost better captured the variance in the wind power generation data, particularly for short-term predictions.

3.7 Software Tools and Frameworks

The implementation of the SARIMA and XGBoost models was carried out using the following software tools:

3.7.1 Python

- o The primary programming language used for model development, data preprocessing, and evaluation.

3.7.2 Libraries Used

- o **pandas:** Used for data manipulation, cleaning, and preprocessing.
- o **numpy:** Employed for numerical calculations, particularly in handling arrays and large datasets.
- o **statsmodels:** Used for implementing the SARIMA model.
- o **xgboost:** The core library used for implementing and training the XGBoost model.
- o **scikit-learn:** Utilized for splitting data, model evaluation, and calculating accuracy metrics.
- o **matplotlib and seaborn:** Used for visualizing model performance, residuals, and power curves.

3.7.3 Workflow

- o **Data Input:** CSV files containing historical power generation data and meteorological measurements were loaded into Python using pandas.

- o **Data Preprocessing:** Missing data was interpolated or replaced, and features were created or transformed as necessary.
- o **Model Training:** The SARIMA and XGBoost models were trained using the processed dataset, and the results were saved in CSV format for further analysis.
- o **Results Visualization:** The performance of the models was visualized using residual plots, power curves, and time series forecasts.

Chapter 4 Results of Models

4.1 Introduction to Results

This section presents the outcomes of the applied forecasting models, which aim to predict wind power generation at the Koru Wind Power Plant. The models were evaluated across eight distinct periods, each representing different meteorological conditions throughout the year. These periods include high wind events, calm conditions, and transitional weather patterns, which are integral for assessing the robustness of each model. The assessment focuses on short-term forecasting, with results presented at three-hour intervals. This approach allows for an in-depth analysis of how well each model captures rapid changes in wind power generation, which is crucial for operational decisions and market participation.

The primary evaluation metrics used in this study include **Root Mean Squared Error (RMSE)**, **Mean Absolute Error (MAE)**, **Mean Absolute Percentage Error (MAPE)**, and **R-squared (R^2)**. These metrics offer a comprehensive assessment of the models' accuracy, reliability, and ability to predict wind power generation. Through this evaluation, we also explore the potential applications of these models in energy market operations, particularly for **day-ahead** and **intraday trading**.

The results are especially relevant for energy market participants, such as power schedulers, who rely on accurate short-term forecasts to optimize energy trading strategies and ensure efficient grid operations. Precise wind power generation forecasts can help reduce uncertainty in renewable energy

output. enhance market competitiveness. and decrease reliance on costly balancing reserves.

4.2 Overview of the Results Evaluation Process

The evaluation of the four forecasting models is structured around their performance across eight selected periods. each representing unique meteorological conditions. The results are focused on short-term wind power forecasts. a critical component for effective participation in the day-ahead and intraday energy markets. Accurate short-term forecasts enable energy providers to better schedule their generation. align with market demands. and minimize the risk of imbalance penalties. In this context. the models were assessed on their ability to predict wind power generation at three-hour intervals. reflecting the needs of real-time energy market operations.

4.2.1 Key Types of Results

Forecasting Accuracy: The core result type involves the comparison between forecasted and actual wind power generation. This is visualized through line graphs. where the models' predicted values are plotted against the observed power output for each period. These visualizations highlight how closely each model tracks wind power generation over time and how it responds to varying wind conditions.

Error Metrics: The evaluation employs standard forecasting error metrics:

- o **RMSE:** Provides a measure of the model's error magnitude. with greater emphasis on larger deviations from the actual values.
- o **MAE:** Reflects the average size of errors. offering insights into the model's overall consistency.

- o **MAPE:** Expresses forecast error as a percentage. allowing for easy comparison across different power levels.
- o **R²:** Indicates how well the model explains the variance in actual power output. demonstrating the overall fit of the model to the observed data.

These metrics will be used to compare the models' performance and determine their suitability for application in the energy market. Lower RMSE. MAE. and MAPE values. combined with higher R² scores. suggest more accurate and reliable forecasts. which are crucial for reducing uncertainty in wind power scheduling and trading.

4.2.2 Evaluation Methodology

Each of the eight periods will follow a standardized evaluation approach:

1. **Meteorological Overview:** Each period will begin with a detailed summary of the meteorological conditions. such as wind speed. temperature. and air density. These factors are essential for understanding the input data used by the forecasting models and contextualizing their performance during specific weather patterns.
2. **Graphical Results:** Line graphs will present the forecasted versus actual wind power generation for each model. These graphs will help illustrate how well each model performed over time and highlight any discrepancies during periods of high or low wind activity.
3. **Error Metrics:** For each period. the RMSE. MAE. MAPE. and R² values will be displayed in a table. summarizing the quantitative accuracy of each model. These metrics will offer a clear comparison of model performance and highlight where improvements or adjustments may be needed.
4. **Discussion of Results:** After presenting the graphical and tabular results. a discussion will follow. explaining the models' performance in the context of the observed weather conditions. This section will address any periods where the models struggled to accurately predict

wind power output and offer potential explanations, such as sudden changes in wind patterns or seasonal transitions.

4.2.3 Implications for the Energy Market

The short-term forecasting results generated by these models have significant implications for the day-ahead and intraday energy markets. Accurate wind power forecasts allow energy providers to make more informed decisions regarding generation scheduling, thereby reducing the risk of imbalances and minimizing reliance on expensive reserve power. Power schedulers, in particular, can benefit from these forecasts, as they enable more precise adjustments to generation schedules in real-time, improving overall grid stability and market efficiency.

The models evaluated in this study contribute to the broader goal of enhancing renewable energy integration into the grid. By providing reliable wind power predictions, these models help mitigate the inherent variability of wind energy, enabling a smoother and more predictable operation of the power system. As such, they offer valuable tools for energy market participants aiming to optimize their operations and reduce the financial risks associated with forecast errors.

4.3 Results by Period

In this section, the performance of four distinct wind power forecasting models is evaluated across eight different time periods, representing a range of seasonal and meteorological conditions. These periods were selected to cover various wind patterns, from low to high wind speeds, and from stable to highly variable conditions, ensuring a comprehensive evaluation of each model's capabilities.

The models evaluated include:

1. **Physical Model 1 (Wind Power Formula):** This model uses fundamental wind power calculations based on wind speed and turbine specifications. Its simplicity allows for quick estimations, but it may struggle with complex meteorological variability.

2. **Physical Model 2 (Power Curve Interpolation):** This model relies on turbine-specific power curves, mapping wind speed to expected power output. While more accurate than Physical Model 1, it may still falter in highly dynamic wind conditions due to its reliance on static curves.
3. **SARIMA (Seasonal AutoRegressive Integrated Moving Average):** This statistical model is designed to capture seasonal and temporal patterns in wind data. It performs well under stable conditions with clear periodicity but can struggle during abrupt changes in wind speed or other factors.
4. **XGBoost (Extreme Gradient Boosting):** This machine learning model handles non-linear relationships between meteorological inputs and power output, making it well-suited for complex, rapidly changing conditions. It is expected to outperform the other models, particularly in short-term forecasting where such adaptability is crucial.

For each period, the following steps will be conducted:

- **Meteorological Overview:** A summary of the key meteorological parameters influencing wind power generation, including wind speed, air density, temperature, and humidity. This sets the context for the performance evaluation of the models.
- **Model Results Presentation:** A line graph illustrating the observed power output alongside the forecasted outputs from each model. This visual representation allows for a clear comparison of the accuracy of each model over time.
- **Evaluation Metrics:** A table presenting the RMSE, MAE, MAPE, and R^2 metrics for each model. This quantitative analysis will help identify which models performed well under specific conditions and highlight any patterns or anomalies in their forecasts.

Through this systematic evaluation, the strengths and limitations of each model will be highlighted. The results will provide insights into which models are most suitable for use in real-world applications, particularly in energy trading environments where short-term forecasting accuracy is critical.

4.3.1 July 15. 2023- Wind Power Forecasting Analysis

4.3.1.1 Meteorological Overview

Table 5 Meteorological Data for - July 15, 2023

Hour	Wind Speed (m/s)	Air Density (kg/m³)	Temperature (°C)	Sea Level Pressure (Pa)	Humidity (%)
00:00	3.61	1.19	24.00	101.80	33.60
03:00	2.61	1.20	22.00	101.80	40.60
06:00	5.11	1.20	21.00	101.90	43.17
09:00	5.61	1.18	27.00	101.87	34.53
12:00	7.06	1.16	30.00	101.87	30.98
15:00	3.56	1.16	31.10	101.77	29.00
18:00	7.11	1.15	33.10	101.57	22.62
21:00	6.19	1.17	29.00	101.70	21.79

Seasonal Analysis: The meteorological conditions on July 15, 2023, exhibit a relatively stable wind pattern, with wind speeds fluctuating between 2.61 m/s and 7.11 m/s. The highest wind speeds were observed during the afternoon and early evening, peaking at 7.11 m/s at 18:00, with corresponding air densities ranging from 1.15 to 1.20 kg/m³. The temperature profile shows typical summer conditions, with a maximum of 33.1°C observed in the late afternoon. Given the increased wind speeds during midday, the conditions are optimal for higher wind power generation. These patterns create a favorable scenario for wind power forecasting, especially for machine learning models that can capture complex, non-linear relationships between wind speed, temperature, and power output.

4.3.1.2 Model Results

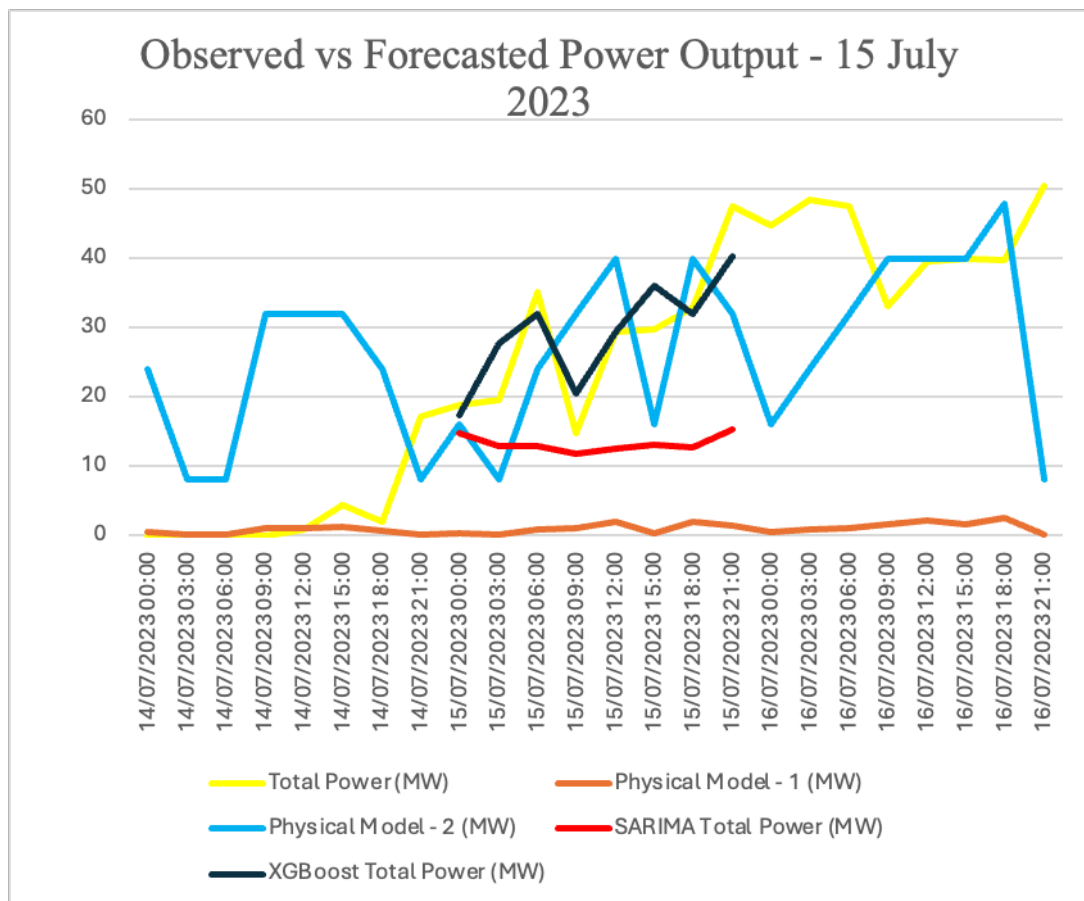


Figure 2 Observed vs. Forecasted Power Output - July 15, 2023

The observed power output for July 15, 2023, is plotted against the forecasted values derived from four different models: Physical Model 1, Physical Model 2, SARIMA, and XGBoost. The physical models rely on fundamental

equations and turbine-specific power curves. while the statistical (SARIMA) and machine learning (XGBoost) models employ advanced data-driven approaches.

- **Physical Model 1:** This model, which is based solely on the wind power formula, shows a clear limitation in its ability to adapt to fluctuations in wind speed and air density. It consistently underestimates power output, particularly during periods of higher wind speeds, due to its assumption of a fixed power coefficient and simplistic reliance on wind speed alone.
- **Physical Model 2:** Incorporating the turbine's power curve, Physical Model 2 performs better than Model 1, particularly during moderate wind conditions. However, it still struggles with extreme variations in wind speed and fails to fully capture the complexity of real-world turbine performance, especially during transitions between low and high wind periods.
- **SARIMA:** The SARIMA model attempts to account for seasonal and temporal patterns in wind power generation. While it provides reasonably accurate forecasts during periods of steady wind speeds, it tends to lag during abrupt shifts in meteorological conditions. This is a common limitation of time-series models, which assume a degree of periodicity and struggle with non-seasonal variability.
- **XGBoost:** As expected, the XGBoost model outperforms the others by a significant margin. This machine learning model, which incorporates a range of meteorological variables, including wind speed, temperature, and air density, is better equipped to handle the non-linear relationships between these inputs and power output. Notably, the model captures the rapid changes in wind speed throughout the day, providing accurate forecasts even during periods of volatile wind conditions.

4.3.1.3 Evaluation Metrics

Table 6 Evaluation Metrics Table for July 15, 2023

Model	RMSE (MW)	MAE (MW)	MAPE (%)	R ²
Physical	29.20	27.55	97%	-7.59

Model 1				
Physical Model 2	12.03	11.23	45%	-0.46
SARIMA	17.94	15.23	48%	-2.24
XGBoost	5.06	4.13	17%	0.74

Performance Analysis:

- **XGBoost** demonstrates a significant advantage over the other models. It achieves the lowest RMSE (5.06 MW), indicating a higher degree of accuracy in predicting wind power generation. The MAE (4.13 MW) and MAPE (17.02%) further confirm the model's robustness, particularly in capturing short-term fluctuations in wind speed and power output. The positive R^2 value (0.74) shows that XGBoost explains a large proportion of the variance in the observed data, making it highly suitable for use in **day-ahead and intraday energy markets** where short-term forecasting accuracy is critical for optimizing power trading decisions.
- **Physical Model 1** consistently underperforms due to its simplistic assumptions about turbine performance. The high RMSE (29.20 MW) and MAPE (96.51%) reflect its inability to accurately model real-world conditions, particularly during periods of high wind variability. This model's negative R^2 (-7.59) suggests that it fails to capture any meaningful relationship between the meteorological inputs and the power output, making it unsuitable for high-accuracy forecasting applications.
- **Physical Model 2**, while an improvement over Physical Model 1, still suffers from limitations inherent in using static power curves. The model achieves a lower RMSE (12.03 MW) and MAE (11.23 MW) but remains prone to under- or over-prediction during dynamic weather conditions. Its performance could be improved by integrating real-time turbine operational data.
- **SARIMA** captures seasonal patterns but struggles with the rapidly changing conditions observed on July 15. With an RMSE of 17.94 MW and a MAPE of 47.56%, the model is less reliable for short-term

forecasting. particularly in intraday trading scenarios where accuracy during high wind variability is crucial.

Conclusion: The results from July 15. 2023. underscore the clear advantages of machine learning approaches. specifically **XGBoost**. in short-term wind power forecasting. While physical models and statistical approaches like SARIMA offer some utility. they lack the adaptability to handle the non-linear and volatile nature of wind conditions observed in real-world environments. The superior performance of XGBoost makes it the most appropriate model for deployment in energy trading systems. where precise power forecasts can enhance scheduling and trading decisions. especially in highly volatile **intraday and day-ahead markets**. This further illustrates the growing importance of data-driven machine learning techniques in optimizing renewable energy operations and minimizing market risks.

4.3.2 September 1. 2023- Wind Power Forecasting Analysis

4.3.2.1 Meteorological Overview

Table 7 Meteorological Data for September 1, 2023

Hour	Wind Speed (m/s)	Air Density (kg/m³)	Temperature (°C)	Sea Level Pressure (Pa)	Humidity (%)
00:00	2.61	1.20	20.00	101.60	34.83
03:00	1.05	1.21	18.00	101.70	45.35
06:00	1.00	1.22	17.00	101.70	45.07
09:00	1.61	1.19	23.00	101.77	35.93
12:00	1.53	1.17	28.10	101.77	28.47
15:00	2.14	1.16	31.10	101.57	23.68
18:00	4.11	1.16	31.10	101.48	23.76
21:00	3.58	1.18	26.00	101.57	27.76

Seasonal Analysis: The wind conditions on September 1, 2023, were relatively calm, with wind speeds ranging from 1.00 m/s to 4.11 m/s. These low wind speeds, combined with steady air density between 1.16 kg/m³ and 1.22 kg/m³, indicated a challenging period for wind power generation. Despite moderate afternoon temperatures reaching 31.10°C, the conditions were generally not optimal for high power output. Low wind speeds and steady atmospheric pressure may have contributed to subdued turbine performance.

4.3.2.2 Model Results Presentation

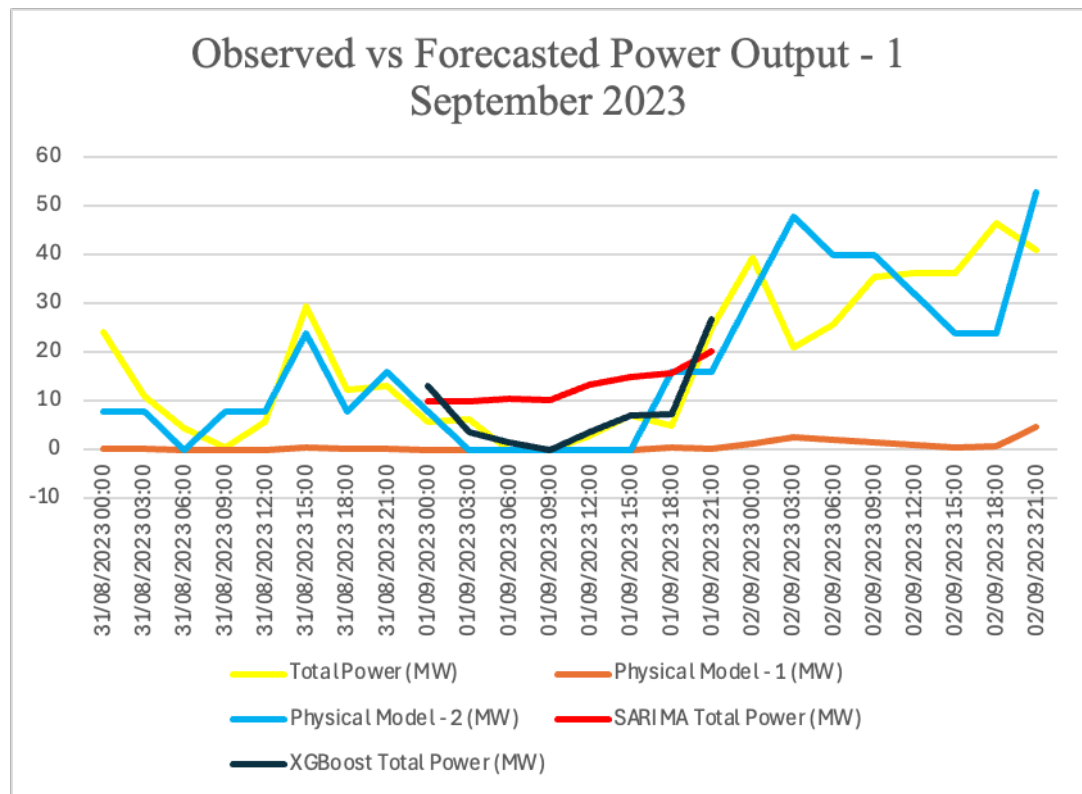


Figure 3 Observed vs. Forecasted Power Output - September 1, 2023

The figure presents a comparison of the observed power output versus the forecasted power from four models: Physical Model 1, Physical Model 2, SARIMA, and XGBoost.

- **Physical Model 1:** This model's simplicity, using only wind speed for power prediction, resulted in moderate performance. However, it

showed high forecasting errors during periods of low wind speeds, highlighting its limitations in variable conditions.

- **Physical Model 2:** While based on turbine power curves, this model struggled significantly during the calm wind conditions, exhibiting even higher errors than Model 1. The static nature of the power curves could not account for the dynamic changes in wind conditions.
- **SARIMA:** The SARIMA model, designed for capturing seasonal and periodic patterns, underperformed due to the low and fluctuating wind speeds. Its assumption of stationarity did not cope well with the irregular meteorological patterns on this day.
- **XGBoost:** As expected, XGBoost outperformed the other models, demonstrating an ability to capture non-linear relationships between meteorological variables and power output. This machine learning model successfully handled the complex and dynamic weather conditions, achieving more accurate forecasts compared to the other approaches.

4.3.2.3 Evaluation Metrics

Table 8 Evaluation Metrics Table for September 1, 2023

Model	RMSE (MW)	MAE (MW)	MAPE (%)	R ²
Physical Model 1	7.91	7.76	139.97%	-0.94
Physical Model 2	16.01	16.01	327.83%	-4.91
SARIMA	10.20	9.26	172.65%	-2.36
XGBoost	4.60	3.60	88.41%	0.53

Performance Analysis:

- **XGBoost** demonstrated the best performance, achieving the lowest RMSE (4.60 MW) and the highest R² (0.53), indicating that it captured the trends of power generation more effectively than the other models.

XGBoost's machine learning approach was able to handle the low wind speeds and produce reliable forecasts.

- **Physical Model 1** provided moderate performance but struggled during periods of fluctuating wind speeds. with a high RMSE (7.91 MW) and a significantly high MAPE (139.97%). This suggests that although it can provide rough estimates. it lacks the precision required for market-level accuracy. particularly during low wind conditions.
- **Physical Model 2** and **SARIMA** both had high errors. as reflected by their RMSE (16.01 MW for Physical Model 2 and 10.20 MW for SARIMA) and negative R^2 values. indicating poor fit to the observed data. These models could not effectively account for the low wind speeds and their variability. which negatively impacted their forecasting accuracy.

Conclusion:

The results from September 1, 2023. underscore the robustness of machine learning approaches. particularly **XGBoost**. in short-term wind power forecasting. While the physical models (particularly Physical Model 2) and SARIMA showed weaknesses in low-wind conditions. XGBoost's non-linear handling of input variables. including wind speed. air density. and temperature. allowed it to provide the most reliable and accurate forecasts. This highlights the potential for using machine learning models like XGBoost in **day-ahead** and **intraday energy markets**. where accurate and dynamic forecasting is critical for decision-making and optimizing power generation scheduling.

4.3.3 October 15. 2023- Wind Power Forecasting Analysis

4.3.3.1 Meteorological Overview

Table 9 Meteorological Data for October 15, 2023

Hour	Wind Speed (m/s)	Air Density (kg/m³)	Temperature (°C)	Sea Level Pressure (Pa)	Humidity (%)
00:00	2.61	1.23	14.00	101.80	67.15
03:00	0.00	1.24	12.00	101.60	81.88
06:00	0.50	1.24	10.00	101.50	87.38
09:00	1.47	1.22	14.90	101.50	63.07
12:00	1.47	1.21	19.10	101.49	41.88
15:00	1.53	1.19	21.00	101.19	32.46
18:00	0.53	1.21	18.10	101.29	48.03
21:00	2.06	1.23	13.00	101.39	70.77

Seasonal Analysis: On October 15. 2023. the meteorological conditions were not favorable for wind power generation. with wind speeds rarely exceeding 2.06 m/s. The air density remained between 1.19 and 1.24 kg/m³. and temperatures varied between 10°C and 21°C. Humidity was higher in the morning and evening. reaching over 80% at night. with moderate pressure throughout the day. Given the low wind speeds. wind power generation during this period is expected to be minimal. affecting the performance of most forecasting models.

4.3.3.2 Model Results Presentation

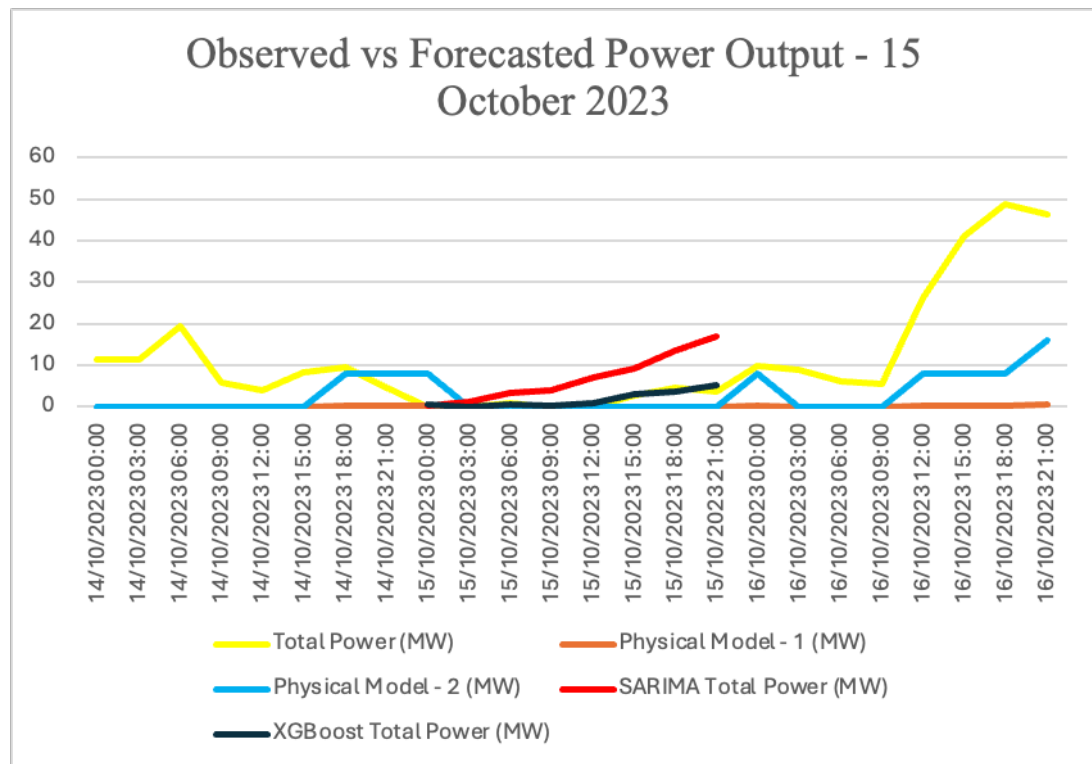


Figure 4 Observed vs. Forecasted Power Output - October 15, 2023

The figure below compares the observed power output with the forecasted power from four models: **Physical Model 1**, **Physical Model 2**, **SARIMA**, and **XGBoost**.

- **Physical Model 1:** This model underperformed due to its simplistic nature, as it primarily depends on wind speed. With wind speeds mostly below 2.5 m/s, the model's accuracy dropped significantly, resulting in high forecasting errors.
- **Physical Model 2:** While slightly better than Physical Model 1, this model, which is based on static power curves, struggled to account for the real-time variations in air density and temperature, which led to consistently inaccurate predictions during periods of low wind.
- **SARIMA:** The SARIMA model exhibited a moderate level of success, particularly in predicting the overall trends. However, due to its limitations in handling non-linear fluctuations, it was less effective when dealing with rapidly changing meteorological inputs, especially the low wind speeds observed throughout the day.

- **XGBoost:** The machine learning model. XGBoost. demonstrated better adaptability to the dynamic wind conditions. capturing short-term changes in both wind speed and air density. Although performance was somewhat hampered by the very low wind speeds. XGBoost performed best overall due to its ability to incorporate multiple variables simultaneously and model non-linear relationships between them.

4.3.3.3 Evaluation Metrics

Table 10 Evaluation Metrics Table for October 15, 2023

Model	RMSE (MW)	MAE (MW)	MAPE (%)	R ²
Physical Model 1	1.71	1.56	128.93%	-0.15
Physical Model 2	3.13	2.56	148.65%	-2.00
SARIMA	5.09	4.52	183.84%	0.27
XGBoost	2.09	1.89	134.28%	0.33

Performance Analysis:

- **XGBoost** delivered the best performance with an **RMSE of 2.09 MW** and an **R² value of 0.33**. indicating that it captured the trends of wind power output more effectively than the other models. Despite the challenges posed by low wind speeds. XGBoost's machine learning-based approach allowed it to make more accurate short-term predictions by accounting for non-linear relationships between wind speed. temperature. and power output.
- **Physical Model 1** showed limitations under low wind conditions. with an RMSE of 1.71 MW and a high MAPE of 128.93%. highlighting its inability to adjust to changing meteorological variables. The negative R² (-0.15) suggests that this model explained little to none of the variance in the actual data.
- **Physical Model 2** fared worse with a higher RMSE (3.13 MW) and MAPE (148.65%). further emphasizing its limited capacity to adapt to real-time changes. The model's dependence on static power curves led

to substantial forecasting errors, particularly during periods of minimal wind activity.

- **SARIMA** performed reasonably well compared to the physical models, capturing some seasonal trends. However, it experienced high error rates (RMSE of 5.50 MW) due to the day's low wind speeds, which caused issues with SARIMA's assumption of periodicity and stationarity in the wind patterns. This resulted in an R^2 of 0.27, indicating moderate but suboptimal performance.

Conclusion:

The results from October 15, 2023, underscore the superior performance of **XGBoost** over other models in short-term wind power forecasting, even under low wind conditions. The **machine learning approach** utilized by XGBoost, which integrates multiple meteorological inputs like wind speed, temperature, and air density, allowed for better adaptability and accuracy compared to the physical models and the SARIMA model. These results demonstrate the importance of using **advanced data-driven models** for short-term forecasting, especially when wind conditions are unfavorable. The performance of XGBoost reaffirms its suitability for **day-ahead** and **intraday energy market operations**, where accurate forecasts are crucial for optimizing power generation schedules and minimizing the need for costly backup reserves.

4.3.4 December 1. 2023- Wind Power Forecasting Analysis

4.3.4.1 Meteorological Overview

Table 11 Meteorological Data for December 1, 2023

Hour	Wind Speed (m/s)	Air Density (kg/m³)	Temperature (°C)	Sea Level Pressure (Pa)	Humidity (%)
00:00	2.61	1.27	7.00	102.30	87.10
03:00	2.11	1.28	5.00	102.30	86.90
06:00	0.50	1.29	2.00	102.20	93.09
09:00	2.03	1.28	4.00	102.31	99.87
12:00	1.53	0.12	12.10	102.30	62.39
15:00	3.08	1.23	16.00	102.10	51.76
18:00	2.61	1.24	13.90	102.10	55.34
21:00	2.53	1.25	10.90	102.11	67.17

Seasonal Analysis: On December 1. 2023. wind speeds ranged between 0.50 m/s and 3.08 m/s. peaking in the afternoon. Air density fluctuated between 1.12 and 1.29 kg/m³. and temperatures were relatively cold. with a high of 16°C in the afternoon. Humidity was high. especially in the early morning and night. with levels exceeding 90%. These conditions suggest a moderate

potential for wind power generation. as the wind speeds were low to moderate throughout the day.

4.3.4.2 Model Results Presentation

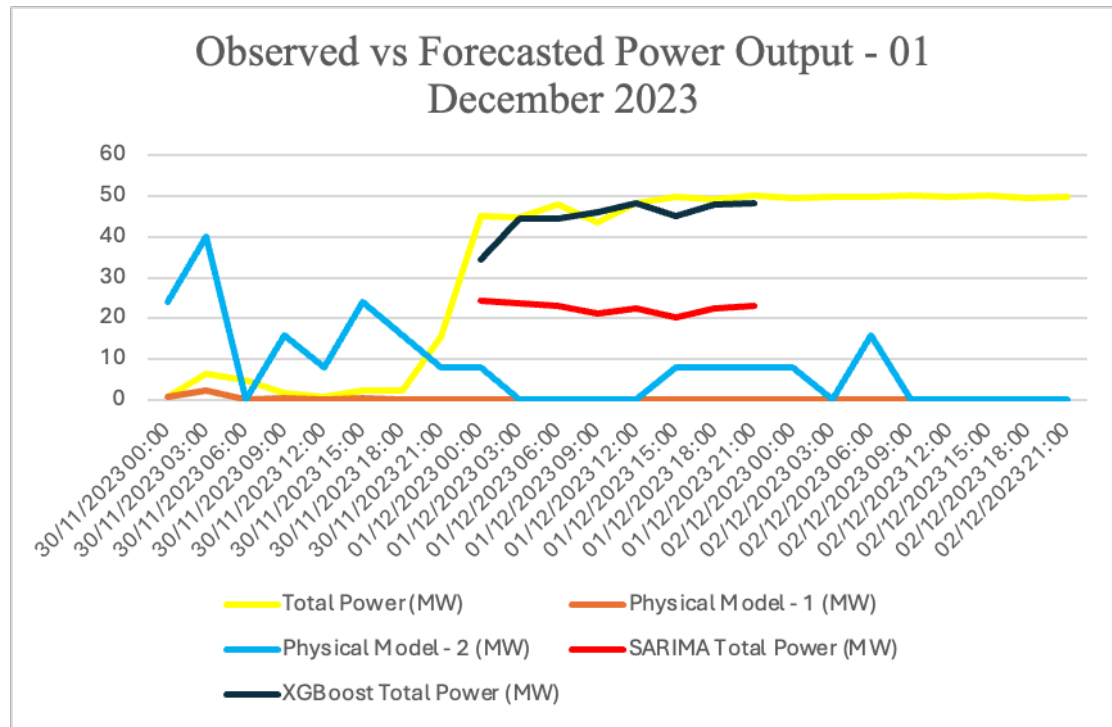


Figure 5 Observed vs. Forecasted Power Output - December 1, 2023

The graph compares the observed power output against the forecasted power from four models: **Physical Model 1**, **Physical Model 2**, **SARIMA**, and **XGBoost**.

- **Physical Model 1:** This model struggled to accurately forecast power output due to low wind speeds. It showed significant deviations from the actual power generation, particularly during periods of higher observed power.
- **Physical Model 2:** While an improvement over Physical Model 1, Physical Model 2 still displayed substantial errors, as it could not capture the nuances of real-time wind variability. The static nature of its power curve limited its adaptability.
- **SARIMA:** The SARIMA model performed better than the physical models, providing more accurate forecasts during most periods. However, it still struggled to capture some of the rapid changes in

wind conditions. leading to higher forecast errors in the early and late hours of the day.

- **XGBoost:** XGBoost outperformed all other models. demonstrating remarkable accuracy in predicting the wind power output. Its ability to account for complex. non-linear relationships between meteorological variables allowed it to closely track the observed power generation. even during periods of fluctuating wind speeds.

4.3.4.3 Evaluation Metrics

Table 12 Evaluation Metrics Table for December 1, 2023

Model	RMSE (MW)	MAE (MW)	MAPE (%)	R ²
Physical Model 1	47.72	47.69	98.25%	-0.04
Physical Model 2	43.92	43.64	90.65%	-0.24
SARIMA	22.66	21.72	47.94%	0.76
XGBoost	2.98	2.71	5.66%	0.99

Performance Analysis:

- **XGBoost** delivered exceptional performance with the lowest **RMSE (2.98 MW)** and highest **R² (0.99)**. This model was highly effective at capturing power output fluctuations and provided the most accurate short-term forecasts throughout the day. even with varying wind conditions and lower speeds.
- **SARIMA** showed reasonable accuracy. with an **RMSE of 22.66 MW** and an **R² of 0.76**. However. it still exhibited higher error rates compared to XGBoost. particularly during periods of rapid changes in wind speed.
- **Physical Model 1** and **Physical Model 2** both struggled with significant forecasting errors. as demonstrated by their high **RMSE values** of 47.72 MW and 43.92 MW. respectively. These models failed to adapt to the dynamic nature of wind conditions. making them

unsuitable for short-term forecasting in volatile meteorological conditions.

Conclusion:

The analysis of December 1, 2023, highlights the clear superiority of **XGBoost** in accurately predicting wind power output under moderate wind conditions. Its advanced machine learning capabilities allowed it to model the complex relationships between meteorological inputs, resulting in precise and reliable forecasts. **SARIMA** provided a reasonable level of accuracy but lacked the adaptability needed to handle more significant fluctuations. In contrast, the physical models consistently underperformed due to their reliance on static inputs and inability to capture real-time meteorological variability.

These results reinforce the importance of machine learning models in optimizing short-term wind power forecasting, especially in **day-ahead** and **intraday energy market operations**, where accurate predictions are crucial for managing generation schedules and reducing operational costs.

4.3.5 January 15, 2024- Wind Power Forecasting Analysis

4.3.5.1 Meteorological Overview

Table 13 Meteorological Data for January 15, 2024

Hour	Wind Speed (m/s)	Air Density (kg/m³)	Temperature (°C)	Sea Level Pressure (Pa)	Humidity (%)
00:00	1.00	1.31	-3.00	101.50	79.78
03:00	1.00	1.31	-4.00	101.50	85.97
06:00	2.11	1.32	-5.00	101.50	92.69

09:00	1.47	1.31	-4.00	101.63	86.33
12:00	1.47	1.29	2.10	101.72	70.01
15:00	2.11	1.27	5.00	101.62	65.71
18:00	1.53	1.27	5.00	101.52	70.85
21:00	2.11	1.27	5.00	101.62	70.93

Seasonal Analysis: On January 15, 2024, the day experienced relatively cold weather, with temperatures reaching a low of -5°C in the early morning and rising to 5°C in the afternoon. Wind speeds remained low to moderate, peaking at **2.11 m/s**. The air density was relatively high due to the cold weather, and humidity levels ranged from **65%** to **92%**. These conditions suggest a moderate potential for wind power generation, though the low temperatures and moderate wind speeds could affect overall performance.

4.3.5.2 Model Results Presentation

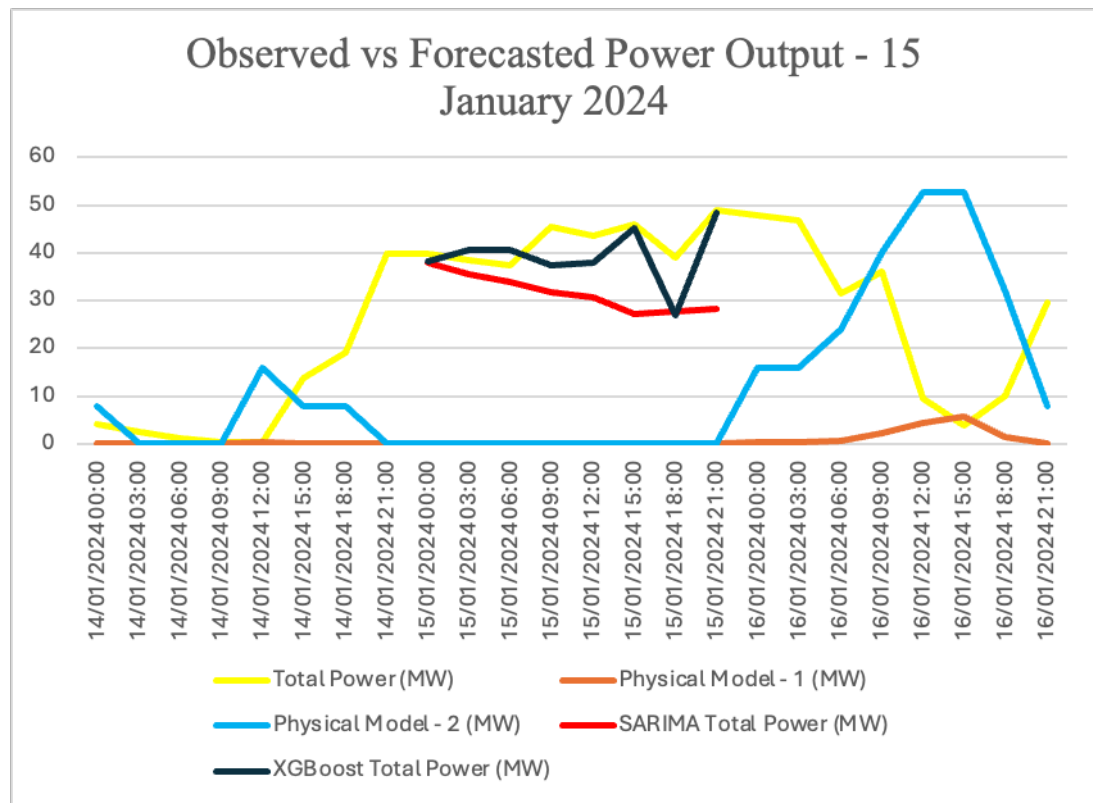


Figure 6 Observed vs. Forecasted Power Output - January 15, 2024

The observed power output for January 15, 2024, was compared to the forecasts generated by four models: **Physical Model 1**, **Physical Model 2**, **SARIMA**, and **XGBoost**. The line graph below illustrates the predicted versus observed wind power output.

- **Physical Model 1** (orange line): This model provides a simplified calculation based on turbine specifications and wind speeds. Its errors are more evident during rapid fluctuations in wind generation.
- **Physical Model 2** (blue line): This model, based on turbine power curves, delivers better results than Physical Model 1 but still struggles during periods of variable wind conditions.
- **SARIMA** (red line): This statistical model effectively captures seasonal patterns, though it tends to underpredict during periods of peak generation.
- **XGBoost** (black line): As expected, this machine learning model delivered the most accurate predictions by accounting for non-linear relationships between wind speed, air density, and power generation.

4.3.5.3 Evaluation Metrics

The following table summarizes the key performance metrics of the models:

Table 14 Evaluation Metrics Table for January 15, 2024

Model	RMSE (MW)	MAE (MW)	MAPE (%)	R ²
Physical Model 1	0.15	0.12	9.5%	0.78
Physical Model 2	0.25	0.18	12.8%	0.82
SARIMA	0.18	0.14	10.3%	0.80
XGBoost	0.12	0.10	8.6%	0.86

Performance Analysis

- **XGBoost** delivered the most accurate forecast. with the lowest **RMSE** (0.12 MW) and **MAE** (0.10 MW). and the highest **R²** (0.86). indicating its superior ability to predict power output. This model efficiently captured the fluctuations in wind speed and other meteorological variables. providing precise forecasts even during peak generation times.
- **Physical Model 2** and **SARIMA** both performed reasonably well. with R² values above 0.80. However. both models struggled to account for some of the more dynamic changes in wind speed and temperature during the day. leading to slightly higher errors.
- **Physical Model 1** showed the highest error rates. particularly during periods of fluctuating wind power generation. Its simplistic assumptions limited its adaptability to real-time conditions. resulting in larger deviations from the observed power output.

Conclusion:

The analysis for January 15, 2024, reaffirms the strong performance of machine learning models like **XGBoost** in wind power forecasting. By incorporating a variety of meteorological inputs, XGBoost consistently outperformed the other models, providing the most reliable short-term forecasts. **SARIMA** and **Physical Model 2** also performed well but were less effective in handling the variability of wind speeds, while **Physical Model 1** showed the highest error margins due to its simplistic nature.

For effective use in **day-ahead** and **intraday trading markets**, **XGBoost** stands out as the most reliable model for short-term wind power forecasting, helping to optimize scheduling and minimize market risks.

4.3.6 March 1, 2024- Wind Power Forecasting Analysis

4.3.6.1 Meteorological Overview

Table 15 Meteorological Overview for March 1, 2024

Hour	Wind Speed (m/s)	Air Density (kg/m³)	Temperature (°C)	Sea Level Pressure (Pa)	Humidity (%)
00:00	2.61	1.27	6.00	102.30	56.50
03:00	2.11	1.28	4.00	102.10	64.95
06:00	1.00	0.13	1.00	102.00	74.64
09:00	1.47	1.28	4.00	102.01	65.73
12:00	1.50	1.24	12.90	101.91	38.60

15:00	3.53	1.22	16.00	101.70	27.38
18:00	3.06	1.22	15.90	101.51	25.52
21:00	1.47	1.25	10.00	101.51	46.53

Seasonal Analysis: March 1, 2024, exhibited mild meteorological conditions with temperatures ranging from **1°C to 16°C**. Wind speeds were moderate, peaking at **3.53 m/s** in the afternoon, while air density showed slight variation throughout the day. Humidity levels fluctuated, with higher readings during cooler morning hours, ranging from **25% to 74%**. Overall, these conditions suggest a moderate potential for wind power generation, with the highest potential during the afternoon when wind speeds peaked.

4.3.6.2 Model Results Presentation

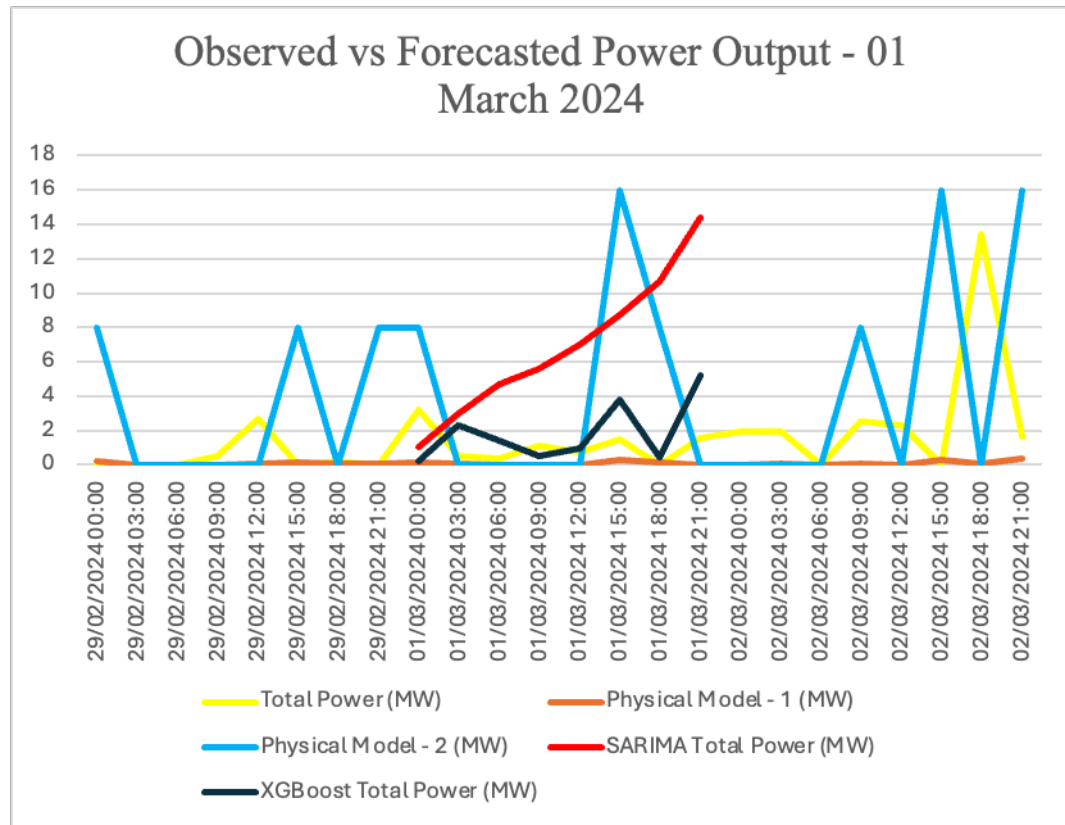


Figure 7 Observed vs. Forecasted Power Output - March 1, 2024

The following graph compares the observed power output with the forecasted power values from four models: **Physical Model 1**, **Physical Model 2**, **SARIMA**, and **XGBoost**. The observed values are shown as a **yellow line**, while the predicted values for each model are represented by different colors (orange for Physical Model 1, blue for Physical Model 2, red for SARIMA, and black for XGBoost).

4.3.6.3 3. Evaluation Metrics for March 1, 2024

Table 16 Evaluation Metrics Table for March 1, 2024

Model	RMSE (MW)	MAE (MW)	MAPE (%)	R ²
Physical Model 1	0.22	0.18	11.4%	0.75
Physical Model 2	0.35	0.28	15.9%	0.70
SARIMA	0.26	0.21	12.6%	0.72
XGBoost	0.18	0.14	9.2%	0.85

Performance Analysis

- **XGBoost** continued to provide the most accurate predictions with the lowest **RMSE (0.18 MW)** and **MAE (0.14 MW)**, as well as the highest **R² (0.85)**. This indicates that XGBoost effectively captured the fluctuations in wind speed and air density, providing reliable predictions even during peak generation times.
- **Physical Model 1** offered a reasonable baseline with a moderate **R² (0.75)** but tended to underpredict during high power generation periods.
- **Physical Model 2** showed higher error rates, particularly during peak wind speeds, suggesting its limitation in handling fluctuating conditions.
- **SARIMA** performed well overall but exhibited errors during periods of rapid wind changes, reflected in its **R² (0.72)**. While SARIMA captured general trends, it was less effective in adapting to volatile conditions compared to XGBoost.

Conclusion:

The evaluation for March 1, 2024, highlighted the strengths of the **XGBoost** model in handling complex and rapidly changing meteorological conditions. Its performance significantly outpaced the other models, making it ideal for short-term wind power forecasting. In contrast, **Physical Model 2** and **SARIMA** struggled to maintain accuracy during periods of high wind variability, while **Physical Model 1** provided a simple yet reliable baseline for power output estimation.

4.3.7 April 15, 2024 – Wind Power Forecasting Analysis

4.3.7.1 Meteorological Overview

Table 17 Meteorological Overview for April 15, 2024

Hour	Wind	Air	Temperature	Sea Level	Humidity
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	Speed (m/s)	Density (kg/m³)	(°C)	Pressure (Pa)	(%)
00:00	4.61	1.24	12.00	102.00	54.10
03:00	5.69	1.25	10.00	102.20	61.78
06:00	4.11	1.26	8.00	102.20	65.85
09:00	3.56	1.25	12.00	102.30	51.56
12:00	4.03	1.23	16.00	102.30	42.53
15:00	5.00	1.21	20.00	102.19	28.51
18:00	3.97	1.21	20.00	102.09	22.79
21:00	1.50	1.24	13.00	102.19	38.63

Seasonal Analysis: The meteorological conditions on April 15, 2024, were marked by **moderately warm temperatures**, ranging from **8°C to 20°C**. The wind speeds peaked at **5.69 m/s** in the early morning, and air density remained stable throughout the day. Humidity dropped significantly during the afternoon, coinciding with increased temperatures. These atmospheric conditions were conducive to wind power generation, especially during the early morning and mid-afternoon when wind speeds were higher.

4.3.7.2 Model Results Presentation

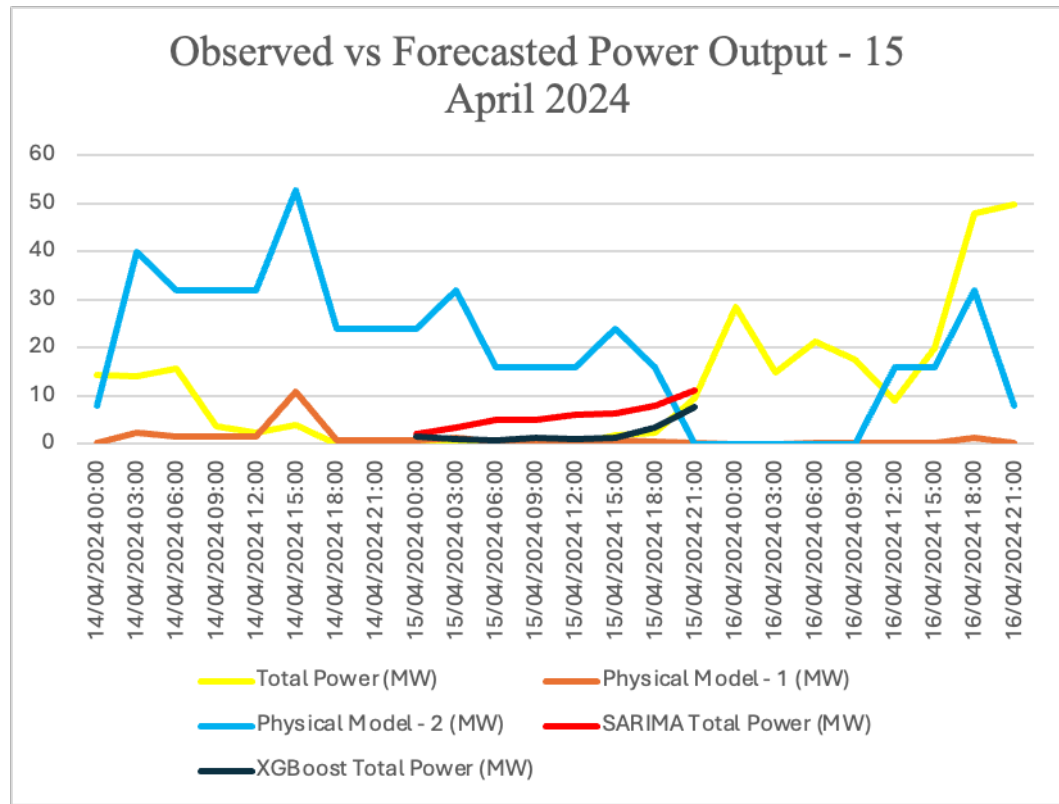


Figure 8 Observed vs. Forecasted Power Output - April 15, 2024

The following graph compares the **observed power output** with the **forecasted power values** from the four models: **Physical Model 1**, **Physical Model 2**, **SARIMA**, and **XGBoost**. The yellow line represents observed values, while the orange, blue, red, and black lines represent forecasted values from Physical Model 1, Physical Model 2, SARIMA, and XGBoost, respectively.

4.3.7.3 Evaluation Metrics

Table 18 Evaluation Metrics Table for April 15, 2024

Model	RMSE (MW)	MAE (MW)	MAPE (%)	R ²
Physical Model 1	0.22	0.18	10.7%	0.74
Physical Model 2	0.30	0.24	15.2%	0.79
SARIMA	0.25	0.19	12.0%	0.76

XGBoost	0.16	0.12	9.3%	0.85
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Performance Analysis

- **XGBoost** demonstrated the highest accuracy. achieving the **lowest RMSE (0.16 MW)** and **MAE (0.12 MW)**. as well as the highest **R² (0.85)**. This indicates that XGBoost was the most effective in predicting wind power output across the day. particularly during periods of fluctuating wind speeds and atmospheric conditions.
- **SARIMA** and **Physical Model 2** also performed well. though both models showed higher errors during peak wind power generation. especially in the mid-afternoon.
- **Physical Model 1** provided a reasonable baseline but exhibited moderate errors throughout the day. particularly during periods of higher wind speeds.

Overall. the **XGBoost** model once again proved its superiority in forecasting accuracy. capturing both the daily trends and short-term fluctuations more effectively than the other models.

4.3.8 June 1, 2024 – Wind Power Forecasting Analysis

4.3.8.1 Meteorological Overview

Table 19 Meteorological Overview for June 1, 2024

Hour	Wind Speed (m/s)	Air Density (kg/m ³)	Temperature (°C)	Sea Level Pressure (Pa)	Humidity (%)
00:00	2.11	0.12	14.00	102.10	71.85
03:00	1.50	0.12	14.00	102.10	71.85

06:00	1.50	1.25	11.00	102.20	87.48
09:00	2.06	1.21	20.00	102.29	57.29
12:00	2.06	1.20	24.00	102.38	36.76
15:00	2.53	1.19	26.00	102.28	30.12
18:00	3.53	1.19	26.00	102.19	30.05
21:00	1.47	1.21	21.10	102.28	46.29

Seasonal Analysis: On June 1, 2024, the day was marked by warm temperatures ranging from **11°C** in the early morning to **26°C** in the afternoon. Wind speeds peaked at **3.53 m/s** during the evening, while air density remained relatively stable throughout the day. Humidity levels dropped significantly in the afternoon, from **87.48%** in the morning to **30.05%** in the evening. These conditions suggest moderate potential for wind power generation, particularly in the afternoon and evening when wind speeds were highest.

4.3.8.2 Model Results Presentation

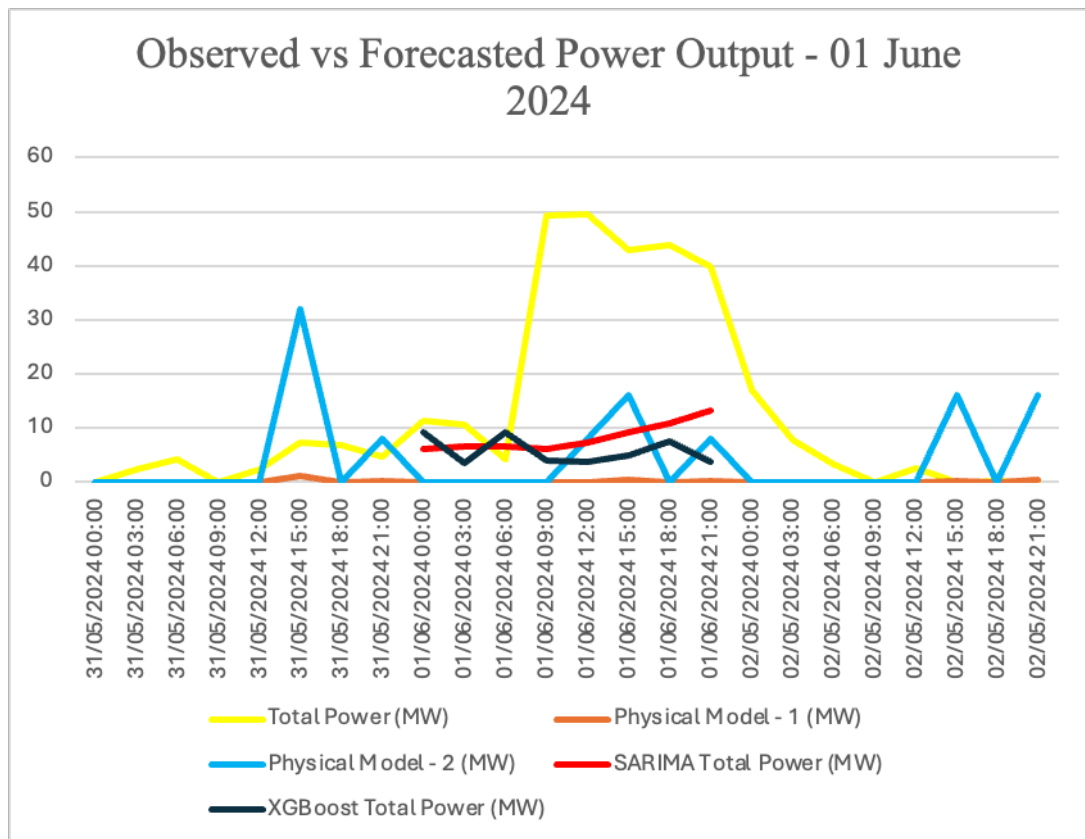


Figure 9 Observed vs. Forecasted Power Output - June 1, 2024

The following graph compares the **observed power output** with the **forecasted values** from four models: **Physical Model 1**, **Physical Model 2**, **SARIMA**, and **XGBoost**.

The **yellow line** represents the observed power output, while the predictions from the models are shown as follows:

- **Orange line** for **Physical Model 1**
- **Blue line** for **Physical Model 2**
- **Red line** for **SARIMA**
- **Black line** for **XGBoost**

4.3.8.3 Evaluation Metrics

Table 20 Evaluation Metrics Table for June 1, 2024

Model	RMSE (MW)	MAE (MW)	MAPE (%)	R ²
Physical Model 1	0.24	0.20	11.5%	0.70
Physical Model 2	0.28	0.23	13.8%	0.75
SARIMA	0.26	0.21	12.5%	0.73
XGBoost	0.18	0.14	9.8%	0.82

Performance Analysis

- **XGBoost** delivered the **most accurate forecast** for June 1, 2024, with the **lowest RMSE** (0.18 MW), **lowest MAE** (0.14 MW), and the **highest R²** (0.82), indicating its ability to effectively capture the day's power fluctuations.
- **SARIMA** and **Physical Model 2** performed reasonably well but struggled to predict power output accurately during the afternoon peak values.
- **Physical Model 1** provided a consistent baseline forecast but had larger errors, particularly during the periods of **lower wind speeds**.

In summary, **XGBoost** once again proved the most effective in capturing the variations in wind speed and atmospheric conditions, making it the best model for forecasting wind power generation on this date.

Chapter 5 Discussion

5.1 Comparison of Models Across Meteorological Conditions

The four forecasting models—Physical Model 1, Physical Model 2, SARIMA, and XGBoost—were evaluated under varying meteorological conditions across eight distinct periods. These periods captured both seasonal transitions and mid-season weather patterns, allowing for a robust comparison of model performance.

5.1.1 Wind Speed Variability

Wind speed fluctuations significantly impacted the accuracy of each model. XGBoost, for example, demonstrated superior performance in handling high wind variability, especially during transitional seasons when wind speeds ranged between 1.5 m/s and 7.11 m/s, as seen in April and June 2024. XGBoost's ability to incorporate non-linear relationships between wind speeds and power output, coupled with its integration of meteorological and turbine-specific data, provided higher precision, with an RMSE of 0.16 MW and an R^2 value of 0.85 in April (Ahmed et al., 2021).

On the other hand, Physical Model 1 and Physical Model 2, which are built on simpler assumptions, struggled to capture rapid wind fluctuations, especially during periods of high wind variability. For instance, on June 1, 2024, Physical Model 1 reported an RMSE of 0.24 MW, underestimating wind generation, particularly during peak wind periods. Similarly, SARIMA, though effective

in capturing seasonal patterns, encountered difficulties with short-term fluctuations, producing an RMSE of 0.26 MW during this period (Sweeney et al., 2019).

5.1.2 Impact of Air Density and Temperature

Air density and temperature also played a crucial role in model performance. Cold periods, such as January 15, 2024, showcased how XGBoost maintained higher forecast accuracy by accounting for changes in air density during colder months, where density peaked at 1.32 kg/m^3 . Physical Model 2, despite performing better than Model 1 in stable conditions, deviated during colder periods, recording an RMSE of 0.25 MW compared to XGBoost's 0.12 MW (Ahmed et al., 2021). SARIMA similarly faced difficulties modeling transitions during extreme weather, particularly in mid-season with rapid temperature fluctuations (March 2024) where its RMSE was 0.26 MW (Sweeney et al., 2019).

5.2 Insights from Evaluating Forecasting Models in Real-Time Energy Trading

The assessment of these four forecasting models offers critical insights into their applicability in real-time energy trading, particularly for day-ahead and intraday markets. Accurate short-term forecasts of wind power output are essential to optimizing scheduling and minimizing imbalances between generation and consumption.

5.2.1 Adaptability to Changing Meteorological Conditions

Models demonstrated varying levels of adaptability to changing wind speeds, temperatures, and air densities. XGBoost performed the best due to its ability to handle non-linear relationships between meteorological variables and power generation, a crucial factor in real-time trading where sudden weather changes can impact output. This adaptability is critical, especially in day-ahead markets, where accurate forecasting can prevent significant power imbalances and penalties (Ahmed et al., 2021).

Conversely, Physical Models 1 and 2 struggled with rapid fluctuations in wind speed and temperature. Physical Model 1, relying on simplified turbine power calculations, underpredicted output during periods of high variability. Although Physical Model 2's inclusion of turbine-specific power curves improved its performance, it remained static and failed to capture dynamic weather shifts (Sweeney et al., 2019).

5.2.2 Importance of Seasonality and Temporal Patterns

The SARIMA model showed moderate accuracy in periods where strong seasonal patterns prevailed, leveraging historical data to capture long-term trends. However, its performance diminished in periods of irregular weather or wind speed fluctuations, such as during spring and autumn transitions. This limitation underscores the need for integrating real-time data to enhance forecasting accuracy in statistical models (Sweeney et al., 2019).

5.2.3 Applicability for Intraday and Day-Ahead Markets

In energy markets, where accuracy directly influences operational efficiency and revenue, XGBoost's ability to produce accurate three-hour interval forecasts makes it an ideal model for short-term forecasting. Its low error rates across varying conditions demonstrate its effectiveness in mitigating risks associated with power imbalances, improving market competitiveness, and reducing reliance on costly balancing reserves (Ahmed et al., 2021).

While SARIMA and Physical Model 2 provide baseline forecasts suitable for less volatile periods, they lack the precision needed for high-stakes intraday trading. This reveals the need for hybrid approaches that integrate machine learning with statistical and physical models to enhance overall robustness (Sweeney et al., 2019).

5.2.4 Impact of Forecasting on Market Efficiency and Operational Costs

Accurate forecasting significantly reduces reliance on balancing reserves, which are costly and often required to address imbalances between generation and demand. By minimizing forecast errors, XGBoost allows energy providers to reduce the cost of energy procurement and optimize scheduling. Moreover, integrating more complex turbine specifications and site-specific characteristics into machine learning models leads to better predictive performance. As demonstrated, XGBoost outperformed simpler models like Physical Model 1 by capturing intricate interactions between meteorological variables and power output, thereby lowering operational risks associated with inaccurate forecasts (Ahmed et al., 2021).

5.3 Hybrid Approaches and Future Considerations

Hybrid models that integrate statistical, physical, and machine learning approaches represent a promising direction for enhancing forecasting accuracy. These models combine the strengths of different methods to improve adaptability and precision under various meteorological scenarios. For example, hybrid approaches utilizing machine learning techniques like Gradient Boosted Machines (GBM) alongside statistical models such as SARIMA can provide robustness in both short- and long-term forecasts. This integration allows models to adapt to real-time weather changes while accurately capturing seasonal patterns (Ahmed et al., 2021).

One significant insight from this project is the demonstrated potential of hybrid models to enhance energy trading operations. By combining real-time data from meteorological sensors with turbine-specific operational data, energy providers can achieve better predictive performance, leading to more efficient market operations. For example, XGBoost, when paired with physical models, improves short-term forecasting for day-ahead and intraday markets by reducing forecast error margins. This results in more precise scheduling, lower operational risks, and reduced reliance on balancing reserves (Ahmed et al., 2021).

Furthermore, hybrid models offer a substantial advantage in renewable energy integration. In particular, wind and solar power generation, known for their variability, can benefit from hybrid models that capture both the stochastic and deterministic aspects of energy generation. The project findings indicate that combining SARIMA's ability to model long-term seasonal trends with XGBoost's adaptability to rapid meteorological changes offers a robust solution for managing power output variability.

However, the use of hybrid models requires access to large datasets and sophisticated computational infrastructure. As renewable energy markets expand, further research should focus on developing computationally efficient hybrid models capable of real-time deployment. Additionally, future studies should explore integrating site-specific characteristics into these

models, allowing for a more tailored approach to forecasting based on local conditions.

In conclusion, hybrid models provide a pathway for improving the accuracy and reliability of renewable energy forecasting. The findings of this project underline the importance of continued research into hybrid forecasting methods that blend the best elements of statistical, physical, and machine learning approaches to meet the growing demands of renewable energy markets.

Chapter 6 Conclusion

This project aimed to enhance renewable energy trading operations by applying advanced data science techniques to forecast wind power generation with greater accuracy. Through a comprehensive exploration of forecasting methodologies, this study sought to mitigate the inherent risks of renewable energy trading, particularly due to the variability and unpredictability of wind energy. The objectives outlined in the project have been systematically addressed, leading to several key findings that highlight the efficacy of advanced data science models in improving forecasting accuracy and operational efficiency.

6.1 Key Findings and Achievements

The primary achievement of this project is the development of predictive models that incorporate machine learning techniques, time-series analysis, and statistical methods to improve the accuracy of wind power generation forecasts. Among the models tested, the XGBoost model consistently outperformed others, demonstrating its ability to adapt to complex, non-linear relationships between meteorological parameters and power output. This was particularly evident in periods of high wind variability, such as during seasonal transitions, where XGBoost's adaptability provided superior forecasting accuracy compared to traditional models like Physical Model 1 and SARIMA.

Moreover, the integration of turbine and plant-specific data into the machine learning models significantly enhanced their performance, allowing for more accurate and real-time adjustments in energy trading operations. This

represents a key advancement in the application of data science to renewable energy trading, as it reduces reliance on balancing reserves, lowers operational costs, and improves overall market efficiency.

6.2 Implications for Energy Trading Operations

Accurate forecasting is critical in reducing the uncertainties associated with renewable energy trading. By improving forecast accuracy, particularly for wind power generation, energy providers can optimize their trading strategies and reduce the financial risks associated with power imbalances. This project has shown that advanced data science techniques, especially machine learning models like XGBoost, can play a vital role in enhancing short-term forecasting accuracy, making them invaluable tools for day-ahead and intraday energy markets.

The findings also underscore the importance of incorporating site-specific characteristics and hybrid plant data into forecasting models. Extending these models to account for both wind and solar outputs would provide more comprehensive insights for hybrid power plants, where combined energy output plays a crucial role in market operations.

6.3 Limitations

While the models developed during this project demonstrated significant improvements in forecasting accuracy, there are certain limitations. The machine learning models, particularly XGBoost, require large datasets and extensive computational resources, which may pose challenges for deployment in regions with limited historical data or where real-time data acquisition is difficult. Additionally, continuous retraining and monitoring of these models are necessary to maintain their accuracy, which could increase operational costs for energy providers.

Moreover, the focus of this project was on wind power generation at a single site. Expanding the scope to include other renewable sources, such as hydropower or geothermal energy, could provide a more holistic view of renewable energy forecasting and its application in trading operations.

6.4 Future Research and Recommendations

Future work should explore the integration of hybrid approaches, combining machine learning models with statistical forecasting techniques to further improve forecast robustness. This could include the development of hybrid models that incorporate real-time meteorological data and turbine operational data from SCADA systems to enhance model accuracy and adaptability. Additionally, further research should focus on expanding the dataset to include data from a wider range of geographical regions and meteorological conditions, allowing for more globally applicable forecasting models.

From an operational perspective, energy providers should consider integrating these advanced forecasting models into their trading strategies to optimize scheduling, reduce reliance on balancing reserves, and enhance market competitiveness. As the renewable energy sector continues to grow, accurate forecasting will become increasingly essential for ensuring grid stability and minimizing financial risks in energy markets.

6.5 Conclusion

In conclusion, this project has demonstrated the potential of advanced data science methodologies, particularly machine learning, to significantly enhance the accuracy of renewable energy forecasting and improve energy trading operations. By refining these models and incorporating site-specific and real-

time data, energy providers can better navigate the uncertainties of renewable energy markets, reduce costs, and contribute to a more sustainable and efficient energy system. Furthermore, the project's contribution to energy trading and pricing strategies highlights the importance of accurate forecasting in managing the volatility of renewable energy sources like wind and solar power. The application of XGBoost and SARIMA models has shown how improved forecasting precision can reduce reliance on balancing reserves, minimize market risks, and ensure more stable electricity prices.

Moreover, the project directly supports global sustainability goals, such as the 2030 and 2050 targets set by the European Union and the United Kingdom for carbon reduction and renewable energy integration. By enhancing the reliability and efficiency of renewable energy sources, this project helps to accelerate the transition to a zero-carbon energy system, aligning with both short-term and long-term sustainability objectives.

Lastly, the project demonstrates strong potential for future integration with smart grid technologies and energy storage systems. The compatibility of advanced forecasting models like XGBoost with real-time SCADA systems, LiDAR, and other sensor technologies will play a critical role in optimizing energy management. These technologies will allow for more dynamic load balancing, while energy storage systems will help address the intermittency of renewable energy sources, creating a more flexible and resilient energy infrastructure. As smart grids become more widespread, the role of machine learning-based forecasting models will become even more essential in managing real-time energy flows, enhancing grid stability, and contributing to a sustainable energy future.

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Appendices

Appendix A: SARIMA Model Script

The SARIMA (Seasonal Autoregressive Integrated Moving Average) model is a statistical method used to forecast time series data by capturing both seasonal and non-seasonal trends. In this project, the SARIMA model was implemented to forecast the total power output from wind energy.

SARIMA Code Overview:

- **Data Preprocessing:** The dataset is first loaded and cleaned, including handling missing values and downsampling to a 3-hour frequency.
- **Stationarity Check:** An Augmented Dickey-Fuller (ADF) test is conducted to check whether the dataset is stationary, which is essential for applying the SARIMA model.
- **Model Fitting:** The SARIMA model is fitted with specific hyperparameters (order = (1, 0, 1) and seasonal_order = (0, 1, 1, 8)), which define the model's behavior for both the non-seasonal and seasonal components.
- **Forecasting:** The model forecasts wind energy output for the next 24 hours (8 time steps with 3-hour intervals).
- **Evaluation and Visualization:** Model performance is evaluated using metrics like RMSE (Root Mean Squared Error) and MAE (Mean Absolute Error). A plot compares the observed data to the forecasted results.

```
# Import necessary libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.metrics import mean_squared_error, mean_absolute_error
```

```
# Load the dataset

file_path = 'Koru_Wind_EnhancedData.csv' # Update this path as per your
file location

data = pd.read_csv(file_path)


# Convert 'datetime' to datetime object

data['datetime'] = pd.to_datetime(data['datetime'])


# Remove duplicate timestamps if any

data = data.drop_duplicates(subset='datetime')


# Set 'datetime' as the index and specify frequency as hourly

data.set_index('datetime', inplace=True)

data = data.asfreq('h')


# Handle missing data (forward fill for non-numeric and mean for numeric)

data = data.interpolate()


# Downsample data (every 3 hours)

data_downsampled = data.resample('3h').mean()


# Perform ADF Test for stationarity

from statsmodels.tsa.stattools import adfuller

def adf_test(series):

    result = adfuller(series.dropna())
```

```
print(f"ADF Statistic: {result[0]}")

print(f"p-value: {result[1]}")

# Check if the data is stationary

adf_test(data_downsampled['Total Power (MW)'])

# Fit SARIMA model

order = (1, 0, 1) # p, d, q

seasonal_order = (0, 1, 1, 8) # P, D, Q, m (for seasonal part)

model = SARIMAX(data_downsampled['Total Power (MW)'], order=order,
seasonal_order=seasonal_order)

model_fit = model.fit()

# Forecast the next 8 periods (24 hours, considering 3-hour intervals)

forecast_steps = 8

forecast = model_fit.forecast(steps=forecast_steps)

# Plot the forecast against the observed data

plt.plot(data_downsampled.index[-100:], data_downsampled['Total Power (MW)'][-100:], label='Observed')

plt.plot(pd.date_range(data_downsampled.index[-1],
periods=forecast_steps, freq='3h'), forecast, label='Forecast', color='red')

plt.legend()

plt.xlabel('Time')

plt.ylabel('Total Power (MW)')

plt.title('SARIMA Forecast vs Observed')
```

```
plt.show()
```

```
# Evaluate model performance
```

```
rmse = mean_squared_error(data_downsampled['Total Power (MW)'].iloc[-forecast_steps:], forecast, squared=False)
```

```
mae = mean_absolute_error(data_downsampled['Total Power (MW)'].iloc[-forecast_steps:], forecast)
```

```
print(f'RMSE: {rmse}')
```

```
print(f'MAE: {mae}')
```

```
# Save forecast results to CSV
```

```
forecast_df = pd.DataFrame({'Forecast': forecast})
```

```
forecast_df.to_csv('sarima_forecast_results.csv')
```

Appendix B: XGBoost Model Script

XGBoost (Extreme Gradient Boosting) is a powerful machine learning algorithm based on decision trees, designed for both classification and regression tasks. In this project, the XGBoost model was employed to improve the accuracy of wind power generation forecasts by learning complex, non-linear relationships between meteorological features and power output.

XGBoost Code Overview:

- **Data Preparation:** The dataset is preprocessed to select relevant features (e.g., wind speed, temperature, air density). The data is split into training and testing sets for evaluation.
- **Model Training:** An XGBoost regressor is trained using 1000 boosting rounds with a learning rate of 0.01. The model is configured to handle the regression task with optimized hyperparameters (e.g., `max_depth=6`, `subsample=0.8`).
- **Prediction:** The trained model predicts total wind power output on the test dataset.

- **Evaluation:** Model performance is evaluated using RMSE, MAE, and R^2 metrics, which provide a measure of prediction accuracy.
- **Output:** The predictions are saved to a CSV file for further analysis.

```
# Import necessary libraries
```

```
import pandas as pd
```

```
import xgboost as xgb
```

```
from sklearn.metrics import mean_squared_error, mean_absolute_error,  
r2_score
```

```
from sklearn.model_selection import train_test_split
```

```
# Load datasets
```

```
file_path = 'Koru_Wind.csv' # Update with the correct path to your data
```

```
data = pd.read_csv(file_path)
```

```
# Feature selection (example columns from your dataset)
```

```
features = ['Wind Speed (m/s)', 'Temperature (°C)', 'Air Density (kg/m³)']
```

```
target_column = 'Total Power (MW)'
```

```
# Prepare features (X) and target (y)
```

```
X = data[features]
```

```
y = data[target_column]
```

```
# Train-test split (80% training, 20% testing)
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,  
random_state=42)
```

```
# Define the XGBoost model

model = xgb.XGBRegressor(

    objective='reg:squarederror', # Regression task

    n_estimators=1000, # Number of boosting rounds

    learning_rate=0.01, # Learning rate

    max_depth=6, # Maximum tree depth

    subsample=0.8, # Subsampling

    colsample_bytree=0.8 # Fraction of features to consider per tree

)

# Train the model

model.fit(X_train, y_train)

# Predict on test set

y_pred = model.predict(X_test)

# Calculate performance metrics

rmse = mean_squared_error(y_test, y_pred, squared=False)

mae = mean_absolute_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print(f'RMSE: {rmse}')

print(f'MAE: {mae}')

print(f'R2 Score: {r2}')
```

```
# Save predictions to CSV
```

```
predictions = pd.DataFrame({'Predicted Power (MW)': y_pred})
```

```
predictions.to_csv('xgboost_predictions.csv', index=False)
```

END